

Prague University of Economics and Business

Faculty of Informatics and Statistics



**FROM STATIC TO DYNAMIC VALUATION:
UNLOCKING THE POTENTIAL OF MONTE
CARLO SIMULATIONS FOR DISCOUNTED
CASH FLOW MODEL**

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Abstract

The thesis explores the use of Monte Carlo simulation to improve the reliability of the DCF method, which is commonly used in business valuation. The object of the valuation is SKODA Auto, the largest car manufacturer in the Czech Republic. The theoretical part of the thesis explains the fundamental concepts of valuation and underlines some limitations of the deterministic approach of the DCF methodology. Monte Carlo simulation is introduced as a tool to approach the variability of model parameters and as a potential tool for generating a range of possible outcomes. In the practical part, the classical DCF model of SKODA is explained, and the Monte Carlo simulation is incorporated into the model using R, the statistical programming language. The simulation produces a sampling distribution of enterprise value probabilities, which, in turn, helps to identify the 95% confidence intervals of the final value. Different scenarios assigned varying probabilities were also tested during the Monte Carlo simulation. The analysis of the results contains measures of central tendency, variability, and shape, as well as the Kernel density estimate. In addition, these findings are further assessed through a finance sensitivity analysis, which aims to identify the most critical components of the entire model. Specifically, the sensitivity analysis revealed that the EBITDA margin is the most critical variable. In summary, the Monte Carlo simulation produced a range of Enterprise value between 92 €mm and 487 €mm, with an average value of 297 €mm.

Keywords

Valuation, DCF, Intrinsic Value, Cash Flow, Monte Carlo, Enterprise Value, Probability Distribution

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Introduction

The valuation of businesses is a crucial part of every investment decision-making process, and the DCF method is the most widely used intrinsic valuation approach in this matter. Nevertheless, it is highly dependent on assumptions regarding the input variables. Therefore, the DCF method can be easily flawed when incorrect assumptions are made in the beginning. This is the reason for incorporating a Monte Carlo simulation into the DCF valuation method, to account for the potential variability of the input variables.

For the thesis, the valuation of SKODA Auto, the largest car manufacturer in the Czech Republic (also the largest company), was chosen as the object of analysis. The company was valued using the DCF method, and the results were then sensitized using the Monte Carlo simulation. In the theoretical part, the fundamentals of business valuation are explained, i.e., the methods used and the pros and cons of each. The DCF approach estimates the value of a business based on the present value of its future cash flows, discounted at a rate reflecting the overall riskiness of the company. However, when projecting the future cash flows, DCF assumes a high degree of accuracy, which might not be the case due to numerous variable factors impacting the forecast, such as economic downturns, shifts in consumer demand, competitive landscape, or any other unpredictable events.

In this context, it is suggested that incorporating Monte Carlo simulation into the DCF model could provide increased reliability of the results. Monte Carlo generates a range of possible outcomes based on the probabilities assigned to input variables. By running multiple simulations, a distribution of possible values can be obtained, which eventually converges to the actual probability distribution. This probabilistic approach provides a more realistic picture of the range of possible values.

In the practical part of the thesis, the DCF valuation model of SKODA is explained step-by-step, while also walking through the Monte Carlo simulation process. First, the process of gathering and forecasting the main financial statements is expounded. Then, the DCF model itself is calculated, using the best financial modeling practices. It is worth noting that the main stress is put on connecting the deterministic and probabilistic approaches to valuation, not solely on the DCF method as such.

The key model input variables that might be subject to uncertainty, such as sales growth rate or EBITDA margin, are identified next. Based on historical financials and industry research, each variable is assigned a probability distribution, including mean value and standard deviation. Subsequently, the Monte Carlo simulation is run to obtain the probability distribution of the enterprise value.

Additionally, to test the robustness of the results obtained through Monte Carlo simulation, different scenarios with varying probabilities assigned to input variables are created, and sensitivity analysis is conducted to identify the most crucial components of the model.

Eventually, the results of the simulation are analyzed, and the consequences of the chosen probability distributions on the result are explained. The analysis includes key descriptive statistics, such as measures of central tendency, measures of variability, and measures of shape. The sampling distribution from the Monte Carlo simulation is also used to derive the 95% confidence intervals of the final value. The probability distribution shows the complete range of values, while the confidence interval suggests that the true values of the company are likely to fall within the range of the 2.5th and 97.5th percentiles, ruling out the extreme values. Finally, while Monte Carlo simulation assumes that the variables are independent (which they are not in reality), it still provides a more comprehensive view of the range of the company value than the sole DCF, by considering the uncertainty and unpredictability of the future.

In conclusion, it is worth noting that all the financial information was collected between August and September 2022, and therefore any new information since then might have not been considered.

THEORETICAL PART

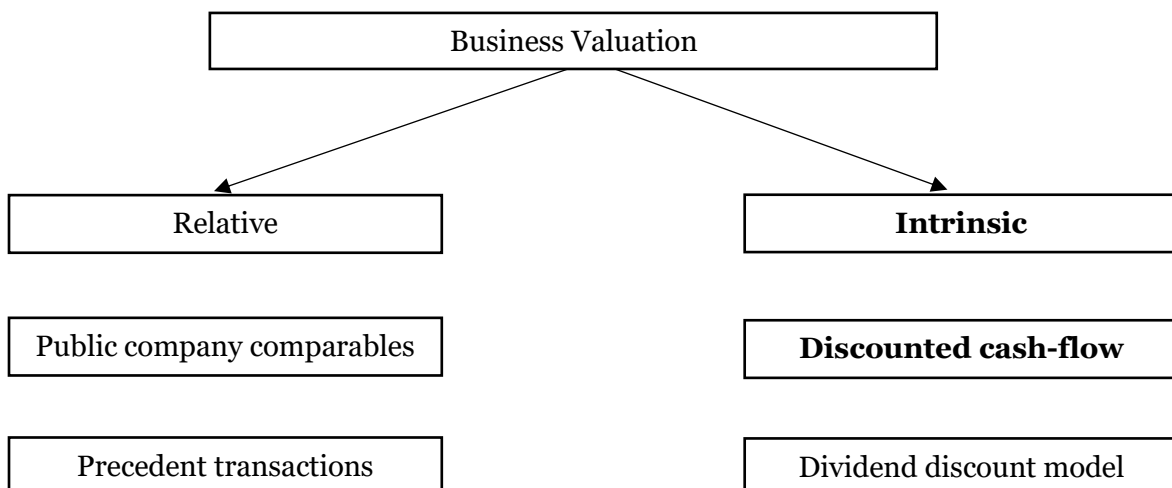
In the theoretical part, the key concepts of valuation are briefly described as well as Monte Carlo simulation and its use case in the discounted cash flow valuation model.

1 Valuation of a business

Multiple approaches exist for valuing a business, with two main concepts being relative valuation and intrinsic valuation.

- 1) *Relative valuation* compares the assessed business with other similar companies using a set of different multiples. This type of valuation is heavily influenced by the current macroeconomic environment, and the multiples may change over time. Common relative valuation methods include EV/EBITDA, EV/Sales, and P/E ratio multiples. Additionally, the Precedent transaction method is commonly used in M&A.
- 2) In contrast, *intrinsic valuation* seeks to determine the intrinsic, or "true," value of the company. It focuses solely on the valued firm and does not consider other companies. The fundamental financials are the most important data required for such a valuation. The most common intrinsic valuation techniques are the discounted cash flow (DCF) analysis and dividend discount model.

While this listing is not complete, these are the most common ways of valuing a business in the world of finance, whether it be investment banking, private equity, or consultancy. It is also best practice to value a business using multiple approaches as a sanity check. In my thesis, I solely focus on the discounted cash-flow method, as it depends on multiple variable factors which will be further examined. (Wall Street Oasis, 2022) (Damodaran, 1994) (Titman & Martin, 2016)



2 DCF Method

DCF is a widely used valuation technique used to estimate the value of an investment or a company based on its future cash flows. In short, DCF estimates the net present value of an investment based on projected future cash flows. There are multiple steps involved in calculating the net present value. (Damodaran, 1994)

DCF steps can be described as following:

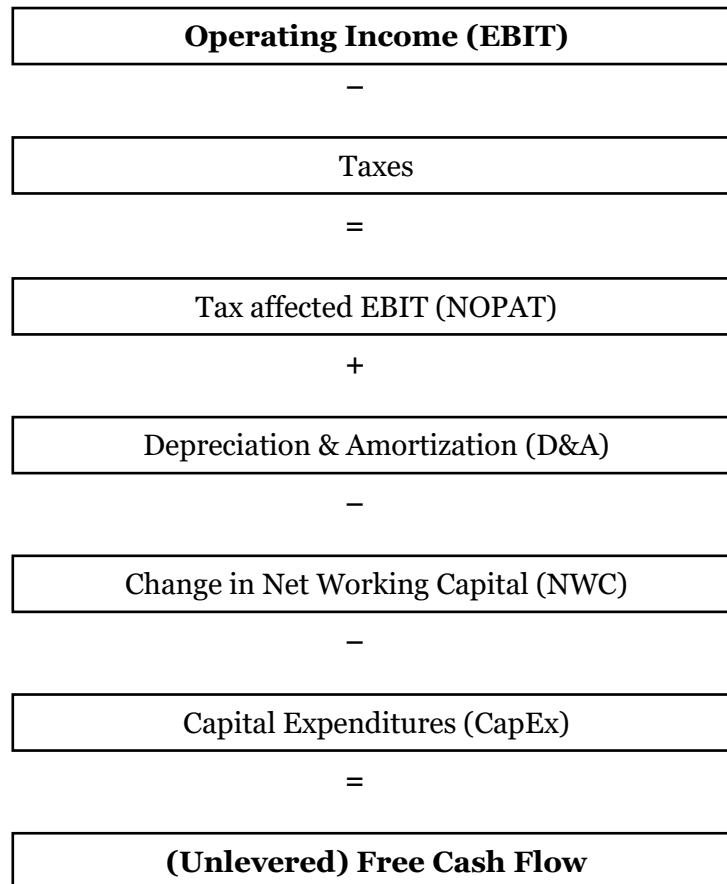
- 1) First, we need to project the free-cash flows of the firm.
- 2) Then, the terminal value must be calculated.
- 3) Choosing adequate *discount rate* (usually in the form of WACC) and discounting both, the projected cash-flows and terminal value to its present value
- 4) After this, we sum the net present value of projected free cash-flows and terminal value to get the Enterprise value
- 5) Finally, the Equity value is calculated by subtracting the net debt (and other adjustments) from the Enterprise value

2.1 Projection of free Cash flows

Before starting the explanation of the steps in free-cash flows calculation, it should be made clear that there are two possible ways of defining free-cash flow, as it can be either levered (LFCF) or unlevered (UFCF). To briefly explain the difference between those two, the unlevered free-cash flow (aka the cash-flow to the firm) is the cash-flow the firm has before debt payments are made. On the other hand, levered free-cash flow (LFCF) is the amount of money left in the firm after all financial obligations have been paid. Levered free cash-flow is sometimes referred to as cash-flow to shareholders because that is exactly the amount of money which is to be divided between them in the form of dividends (or reinvested back to the business). Since in DCF analysis, the Unlevered free free-cash flow approach is mostly used as we aim to calculate the total enterprise value. (Flores, 2023)

Now that the difference between Levered and Unlevered free cash-flow is known, the next step is to explain how these cash-flows can be projected. The first step is to define the projection period, which typically varies between 5 to 10 years depending on the maturity and stability of the business. In this case, a 5-year period was chosen as the company being valued is considered mature and relatively predictable. If a young company operating in a less predictable industry were being valued, a longer period would be chosen for the explicit projection period.

The Unlevered free cash-flow (UFCF) is calculated by subtracting taxes from the Operating Income (EBIT), adding back Depreciation & Amortization, and then subtracting the change in Net Working Capital and Capital Expenditures. The concept of the funnel from EBIT to FCF can be better understood through the following scheme:



The derivation of free cash-flow requires forecasting each item of the Income Statement and Balance Sheet on a line-by-line basis. Consequently, the results of FCF are susceptible to a variety of assumptions, which include the following:

Income Statement items:

- a) Revenue growth
- b) Margins (Gross Margin, EBITDA, EBIT)

Balance Sheet items:

- a) CapEx
- b) Change in NWC

However, detailed projection of all the Three statements (P&L, BS, CF) should be done when robust and reliable valuation model is the goal of the analysis. (Vernimmen et al. 2022)

2.2 Calculation of the Terminal value

The value of the company after the explicitly projected period is represented by the Terminal value. In DCF analysis, we forecast cash-flows only a few years into the future because forecasting beyond 10 years is too unpredictable. Thus, we need to account for the remaining enterprise value, assuming that the company will operate perpetually. During this phase, the cash-flow generating process tends to be stable and normalized, resulting in

slowed down and sustainable growth. There are two widely accepted approaches to calculating Terminal value: the Perpetuity growth method and the Exit multiple method. (Damodaran, 1994)

2.2.1 Perpetuity growth method

The perpetuity growth method, which is also known as the Gordon growth method, involves assuming that the company will grow at a stable and sustainable rate to perpetuity. The perpetuity growth rate must be realistic to achieve accurate results. There are two primary economic indicators that can be used as a reference point when selecting the appropriate growth rate. The first is the GDP growth rate of the relevant country, which serves as an upper limit. Exceeding GDP growth in the long run would imply that the company would eventually dominate the entire economy. Second, the inflation rate is sometimes used as a reliable guideline when searching for the perpetuity growth rate, but it should not exceed the long-term inflation rate. Once all variables have been decided upon, the terminal value can be calculated as follows:

$$TV = \frac{FCFn * (1 + g)}{r - g}$$

Where:

FCFn = Unlevered Free Cash-flow in the final year of the projection

g = perpetuity growth rate

r = discount rate

2.2.2 Exit multiple method

Although the exit multiple method utilizes an exogenous factor, the multiple, to calculate the terminal value, it is still widely used in the financial industry due to its simplicity compared to the perpetuity growth method. First, a reasonable EV/EBITDA multiple (or range) is identified based on comparable companies operating within the same industry. Next, the selected multiple is multiplied by the estimated EBITDA for the final year of projection. The resulting value represents the Enterprise value of the company at the end of the forecasted period. In case there is uncertainty regarding the appropriate multiple to be used, selecting a likely range can be a solution. In this scenario, the Enterprise value for the upper and lower limits is calculated as described above, providing insight into how the value could fluctuate based on the multiple input. (Damodaran, 1994)

$$TV = EV/EBITDA \text{ multiple} * EBITDAn$$

2.2.3 Reconciling the two methods

A good practice and a useful sanity check is to combine the two methods. When combined, it is important to ensure that the respective growth rate and EV/EBITDA multiple remain the same, or at least similar, to the initial input.

EV /EBITDA multiple to check for the Perpetuity growth rate formula:

$$g = \frac{EBITDA_n * \frac{EV}{EBITDA} \text{ multiple} * r - FCF_n}{EBITDA_n * \frac{EV}{EBITDA} \text{ multiple} + FCF_n}$$

Perpetuity method to check for the EV/EBITDA multiple formula:

$$\frac{EV}{EBITDA} \text{ multiple} = \frac{FCF_n * (1 + g)}{EBITDA_n * (r - g)}$$

2.3 Calculating the discount rate (WACC)

Now that both the projected Cash-flows and Terminal value have been calculated, they need to be discounted back to their net present value. To do this, the appropriate discount rate needs to be known. In the case of DCF analysis, the Weighted Average Cost of Capital (WACC) is used. Technically speaking, the WACC is the required rate of return for the firm given the risks to its investors (shareholders & credit holders). This concept is especially important because, since the Unlevered free cash-flow is used, which is attributable to all investors, the cost of both debt and equity must be considered. (Vernimmen et al. 2022)

The cost of alternative investment opportunities available to the firm's investors, given the riskiness of the investment into the company, can explain the required rate of return. If there are investments in the market where investors can obtain X% for the same level of risk, equally risky companies are compelled to offer a similar return, or else they will not receive funding as they would be less attractive to investors. (Vernimmen et al., 2022) (Damodaran, 1994)

The WACC is dependent on the capital structure of the assessed firm, with equity being typically more expensive than debt. However, if the firm employs too much leverage, it becomes riskier, which increases the cost of capital. Hence, a well-balanced mix of debt and equity is usually preferred by most firms.

Rarely is financing obtained through one source only, and the calculation of the WACC becomes more complicated as a result. Each source of capital has its own cost that is weighted based on its share in the company's capital structure. Therefore, the WACC is calculated as the weighted average of the cost of equity and the cost of debt, with weights being the respective shares of equity and debt in the company's capital structure. Additionally, other sources of capital such as preferred stock and minority interests may also need to be considered in the calculation.

Most often, since high WACC signalizes a risky investment, the lower the WACC, the better. The riskier the investment, the more compensation is required by investors. (Anon., 2023)

WACC formula:

$$WACC = \left(\frac{D}{D + E}\right) * K_D * (1 - T) + \left(\frac{E}{D + E}\right) * K_E$$

Where:

K_E = Cost of Equity

K_D = Cost of Debt

E = Market value of Equity (Market capitalization)

D = Book value of Debt

T = Marginal tax rate

2.3.1 Cost of Equity

Cost of Equity is a return a company would theoretically pay to its shareholders for the risk they undertake by investing into the company. The traditional approach for calculating the Cost of Equity is via the Capital Asset Pricing Model (CAPM). It is determined by the following equation:

$$K_E = R_f + \beta(E(R_m) - R_f)$$

Where:

K_E = Cost of Equity

R_f = Risk-free rate

$E(R_m)$ = Expected market return

$(E(R_m) - R_f)$ = Equity risk premium

β = Beta

The Cost of Equity is determined by the CAPM formula, which is based on three factors. First, the Risk-free rate represents the minimum expected return that investors are willing to accept in exchange for zero financial risk. Typically, the yield on short-term U.S. Treasury bills serves as the benchmark for the Risk-free rate. However, depending on the country of operation, a premium may be added based on the country's credit score. The Equity risk premium represents the average excess return earned by investing in equity markets over the Risk-free rate. It is crucial to use the current market spread, as the spread between $E(R_m)$ and R_f varies every year. Beta is a measure of the security's volatility compared to the market as a whole, with the S&P 500 index being the standard. Securities with Beta

greater than 1.0 tend to move with more momentum than the market, regardless of direction. In other words, the market value of the security swings more than that of the market. Mathematically, Beta can be calculated as the covariance between the market return and the return of the individual security, which reflects how movements in a security's returns correlate with market returns, divided by the variance of the market. (Damodaran, 1994)

$$\beta = \frac{Cov(R_E, R_M)}{Var(R_M)}$$

Where:

R_E = Return of the selected security

R_M = Market return

2.3.2 Cost of Debt

The Cost of debt represents the effective interest rate that the company would have to pay on all its debt-like liabilities, primarily loans and bonds. Other forms of debt financing may include revolving facilities or mezzanine financing. In essence, the cost of debt can be simply interpreted as the weighted average interest rate of the entire company. Due to the tax-deductible nature of debt, the cost of debt is multiplied by the corresponding tax discount (1 - Tax rate) for the purpose of calculating WACC. (Damodaran, 1994) (Vernimmen et al., 2022)

2.4 Discounting back UFCF and TV

With the knowledge of WACC, the projected figures of Unlevered free cash-flow and Terminal value (if Perpetuity growth method is used) can be discounted back. It is important to note that the WACC of the projection period may be slightly different from the one used for the Perpetuity part of DCF, where the future capital structure of the firm may need to be considered.

Sum of the discounted figures of projected cash-flows and terminal value determine the Enterprise value of the company. (Damodaran, 1994)

2.5 Enterprise value to Equity value

Apart from the Enterprise value, which is already known at this point, the sole Equity value is the figure of interest when comparing stocks. However, when valuing the business, the Enterprise value is more commonly used as companies are made more comparable by not considering their capital structure.

To clarify the difference between these two, the Enterprise value is the ultimate value of the entire business, not considering its capital structure. On the other hand, Equity value is the part of the business attributable to its shareholders only.

To derive the Equity value from the Enterprise value, a few adjustments need to be made to the balance sheet. Firstly, the net debt (debt less cash) is subtracted from the Enterprise value. Secondly, any minority interests and preferred shares (if applicable) are also subtracted. This bridge can be visually represented by a waterfall chart.

3 Uncertainty of DCF outputs

Despite being the most widely used and reliable valuation technique in the financial industry, the DCF analysis has several limitations. One of the main weaknesses of the DCF is that it assumes the predictability of the future performance of the company, particularly the cash-flow generation, which in reality is often not the case. There are various inputs that go into the DCF model, and to compensate for this, the output of the DCF should be represented by a range of values, rather than a single figure. For example, while the cash-flow projections for the first one to two years may be easy to estimate, estimating the cash-flows five or ten years from now may be difficult due to the uncertainty about the future performance of the company. There are many factors that can influence the results, so it is almost necessary to conduct sensitivity analysis. (Edner & Paulsson, 2013) (French & Gabrielli, 2005)

The most variable inputs affecting the final figure (be it enterprise or equity value) are:

- a) Free-cash flow annual growth rate
- b) Gross margin for the projection period
- c) EBITDA margin
- d) CapEx for the projection period (as % of Revenues)
- e) Perpetuity growth rate (or EV/EBITDA multiple when exit multiple used)
- f) WACC

The industry standard is to run sensitivity analysis in Excel using the sensitivity table function. However, this functionality only allows us to follow two variables at once. While this feature might be useful for some investment considerations, the economic environment in which companies operate is far more complex. DCF itself is an extremely deterministic way of evaluating investments. (French & Gabrielli, 2005) (Ali et al., 2010)

In my research, I delve into how uncertainty and probability can be incorporated into the widely used, but deterministic DCF approach. To address the variability of DCF inputs, I use Monte Carlo, an econometric simulation based on random sampling.

3.1 Monte Carlo simulation

To start, Monte Carlo simulation is used to model the probability of different outcomes in a process that cannot be easily predicted due to the intervention of random variables. It is a technique used to understand the impact of risk and uncertainty. (Kenton, 2023)

The theory behind Monte Carlo simulation is simple: it generates thousands of random samples of uncertain variables. Each time a new sample is generated, a new random number is assigned to the uncertain variable. By generating a large number of random samples, a normal distribution is formed. This provides us with an idea of how probable different outcomes are, with the most likely results being located in the middle of the normal

distribution bell curve. In summary, thanks to the law of large numbers, we can assign a probability distribution to otherwise uncertain variables. However, it is important to be aware that Monte Carlo simulation is run under the *ceteris paribus* condition, meaning that anything that is not incorporated into the model is not reflected in the results (such as macro trends or cyclical factors). (Kenton, 2023)

Basically, there are 4 main steps every Monte Carlo simulation tends to follow:

- 1) Definition of a domain of possible inputs
- 2) Generation of random inputs from a probability distribution
- 3) Performing a deterministic computation on the inputs (known equations)
- 4) Summarizing the results, plotting the probability distribution

In the practical part of this thesis, I will go over all of those steps to find the best possible estimate of the range at which the company should be valued. (Kenton, 2023)

3.2 Incorporation of Monte Carlo into the DCF model

As already described above, the Monte Carlo simulation is incorporated into the DCF model to account for the uncertainty of the model inputs. In the original DCF approach, the key drivers are inputted into the model as hard-coded values (e.g., annual sales growth of 3%). However, with the Monte Carlo extension, those variables are assigned probability distributions based on their potential upsides or downsides and empirical research. It is often the case that some variables tend to follow certain historical patterns. (Shest, 2022)

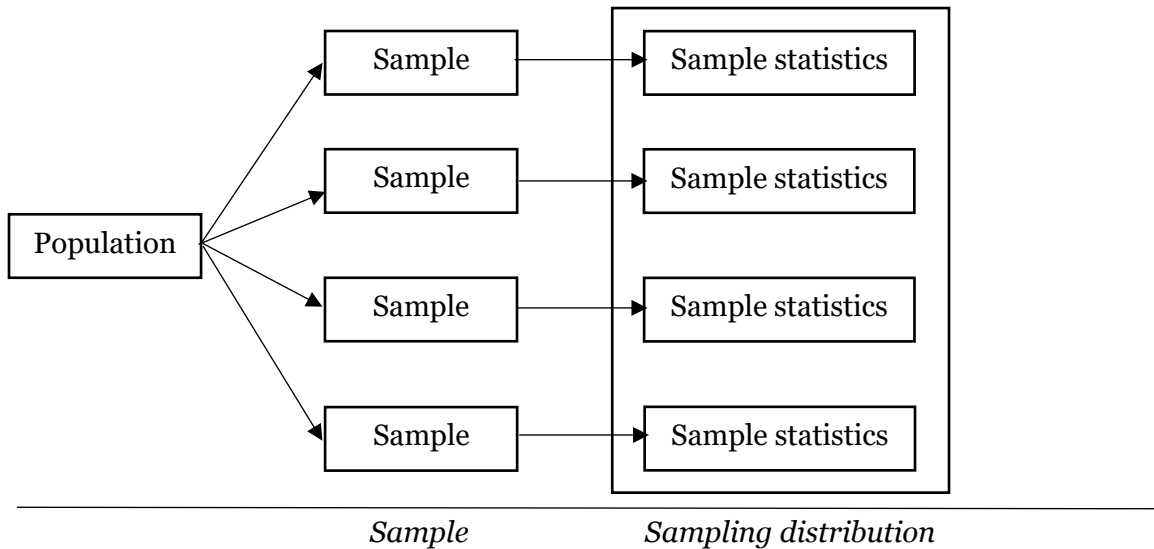
It is not necessary to model all the variables as probabilistic. If we know that some variables are easy to predict (based on comparable companies, historical results, etc.) or will not significantly affect the result, it might be a better idea to keep the input deterministic. Once all the key variables have been assigned the best-fitting probability distribution, the Monte Carlo simulation can begin. Due to its computational demand, the simulation is typically run in more sophisticated software (such as Python, R, etc.).

First, a sample is drawn from the distributions that has been defined for each variable. Then, the final valuation is calculated using the standard DCF formula. The resulting valuation is stored, and the process repeats until the required number of simulations have been run (I typically run 10,000 repetitions). Finally, the stored values of each simulation run are used to generate the required probability distribution of valuation outcomes (i.e., enterprise value). The obtained sampling distribution helps to better understand the most probable valuation, as well as the potential downside, upside, and other related factors. (Cole, 2020) (Shest, 2022) (Edner & Paulsson, 2013) (Ali et al., 2010)

3.2.1 Sampling vs Sample distribution

In the previous section, the concept of a sampling distribution was mentioned. It is important to further explain this concept, as the words 'sample' and 'sampling' can be confusing. To begin, it is necessary to distinguish between the population distribution and

the sampling distribution. This distinction is critical when working with concepts such as the central limit theorem, as well as the standard deviation or standard error. (Alizadeh, 2021)



Statistical inference is a critical concept that deals with the generalization of the obtained data. The assumption is that the data should come from a randomized process. Monte Carlo simulation fulfills this condition as it generates thousands of random samples. Although the estimates will differ in each sample, there are clear values of parameters that accurately describe the relationships in the entire population, such as the true intrinsic value of the company. Therefore, our samples are not exact and contain some degree of error, known as the sampling error. (Zouhar, 2022)

In the case of Monte Carlo simulation, the magnitude of this sampling error depends on how many iterations we run. The central limit theorem comes into play here, as it states that the distribution of a sample variable approximates a normal distribution as the sample size becomes larger, assuming that all samples are identical in size and that the population's actual distribution shape is irrelevant. Therefore, the larger the number of iterations, the more accurate the estimation of the true population distribution will be. However, it is important to keep in mind that even with a large number of iterations, there will always be some degree of uncertainty due to the randomness of the process. (Ganti, 2022)

3.3 Probability distributions

“The Monte Carlo simulation is based on the probability events to occur.” (Edner & Paulsson, 2013). In Monte Carlo simulation, a variety of probability distributions can be used to draw the input variables used to model a system. Therefore, understanding what probability theory is and how random variables can be distributed is necessary for further work. In this section, the distributions that are most suitable for the purposes of a Monte Carlo-simulated DCF model will be explained. The four main distributions that will be described are the Normal distribution, Triangular distribution, Log-normal distribution,

and Uniform distribution. These four types of distributions have previously been used in Monte Carlo simulations and, based on experience in the finance and consulting industries, are commonly used to model real-life situations that businesses might face in the real world. (Edner & Paulsson, 2013).

“Probability distribution is a mathematical function that describes the probability of different possible values of a variable.” (Turney, 2022) There are two categories of distributions that can be used to explain the behaviour of random variables: discrete and continuous. A discrete distribution includes only variables that can take on specific values within a domain, such as the number of heads in a series of coin flips. The overall probability of a specific outcome can be found by summing the probabilities of all the possible discrete outcomes. In contrast, a continuous distribution is used to model variables that can take on any value within a specified domain. In a continuous distribution, the probability of a specific outcome is typically zero, and the probability of a range of outcomes is found by calculating the area under the probability density function within that range. (Turney, 2022)

Most variables in DCF (and business models overall) are associated with either inflow or outflow of cash/money. Technically, money can only be divided into a limited number of fractions. For instance, one euro can only be divided up to cents, which means that money-related variables should be considered discrete. However, if the values are large and the gaps between possible values are small enough, it is often possible to use a continuous distribution as an estimate for these discrete variables. This is the case for large expenditures, which is the focus of this model. As this study mainly focuses on large values, the various probability distribution methods discussed are all continuous. (Edner & Paulsson, 2013) (Hayes, 2023)

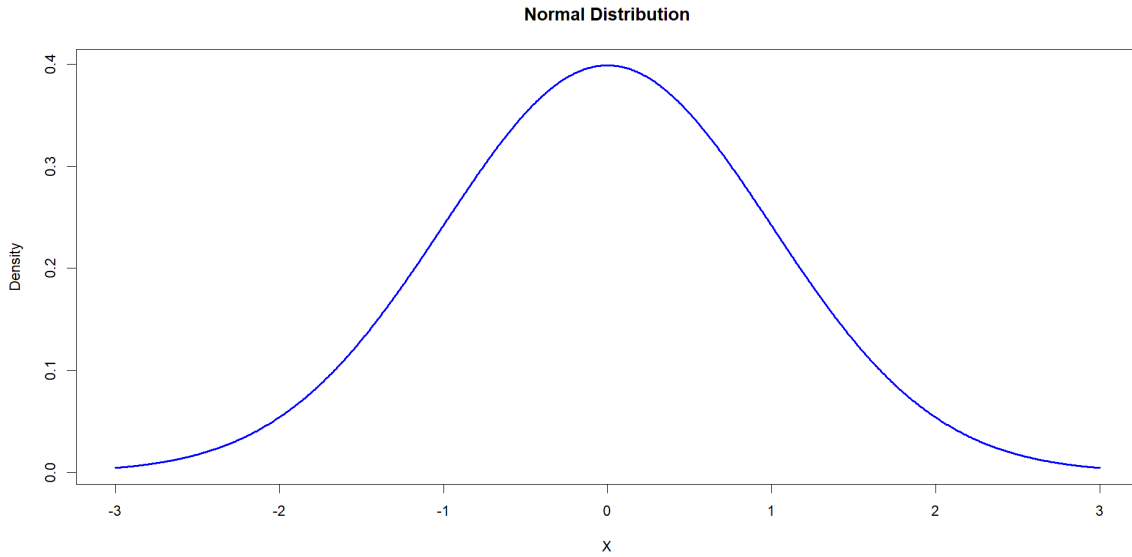
3.3.1 Normal distribution

The normal distribution (also called Gaussian) is one of the most widely used distributions in science, as well as in other fields. It is characterized by two key parameters: the mean (the average or centre of the curve) and the standard deviation (the spread of the curve). The normal distribution is symmetrical about its mean, and its shape is characterized by a peak at the mean and a gradual decay towards the tails. The standard normal distribution is a special case of the normal distribution with a mean of zero and a standard deviation of one.

The normal distribution has several important properties. One of the most well-known is the central limit theorem, which states that averages calculated from independent, identically distributed random variables have approximately normal distributions, regardless of the type of distribution from which the variables are sampled, provided that the distribution has finite variance. (Ross, 2017).

In conclusion, the normal distribution is a widely used and important concept in statistics and probability, with many applications in a variety of fields.

Fig. 3.1 Normal distribution example (author's construction)



The equation below represents the PDF (probability distribution function), where μ is the mean of the distribution and σ is the standard deviation, which describes the spread of the distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

In the thesis, due to its widespread use in the world of science (finance being no exemption), the normal distribution will be found most frequently.

3.3.2 Triangular distribution

A triangular distribution is a continuous probability distribution with a probability density function shaped like a triangle. It is defined by three values: the minimum value a , the maximum value b , and the peak value c . This is handy in real-life situations, as we can often estimate the maximum and minimum values, as well as the most likely outcome, even if we don't know the mean and standard deviation. (Ward Petty & Dye, 2013) Since the triangular distribution has both upper and lower boundaries, it can help to avoid extreme values, as compared to other distributions.

In finance, as well as in other business areas, the triangular distribution finds its use in situations where there is uncertainty about which trend or distribution a variable might follow, but we do have information about the maximum, minimum, and most likely values. The probability density function (PDF) of the triangular distribution can be described by the following equations:

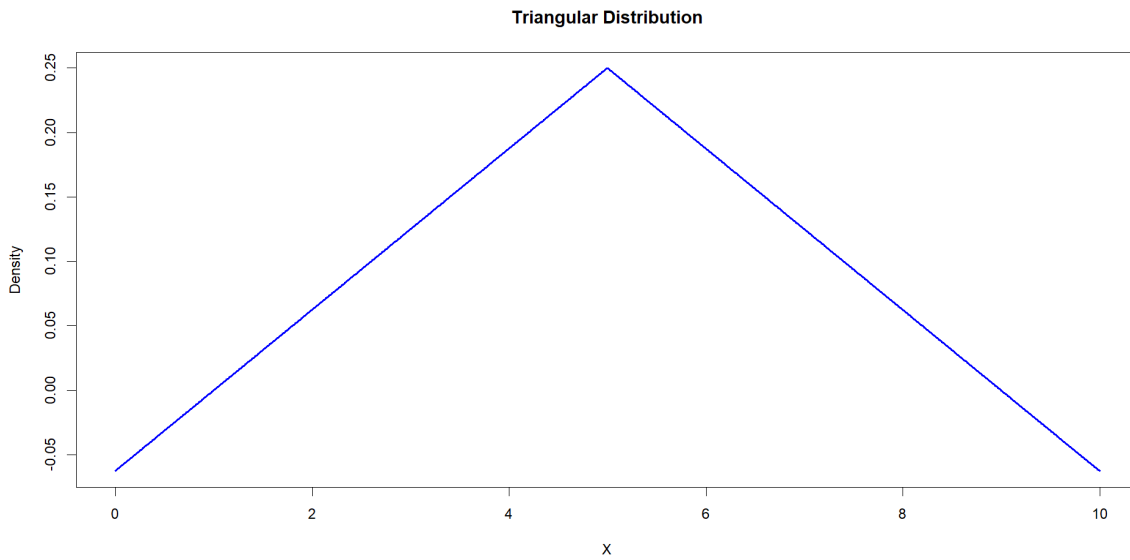
$$f(x) = \frac{2(x - a)}{(b - a)(c - a)} \quad \text{for } a \leq x \leq c$$

$$f(x) = \frac{2(b - x)}{(b - a)(b - c)} \quad \text{for } c \leq x \leq b$$

$$f(x) = 0 \quad \text{otherwise}$$

Where probability for x to occur is defined by its minimum value (a), its maximum value (b), and its mode (c), which is the most likely value to occur. (Ward Petty & Dye, 2013)

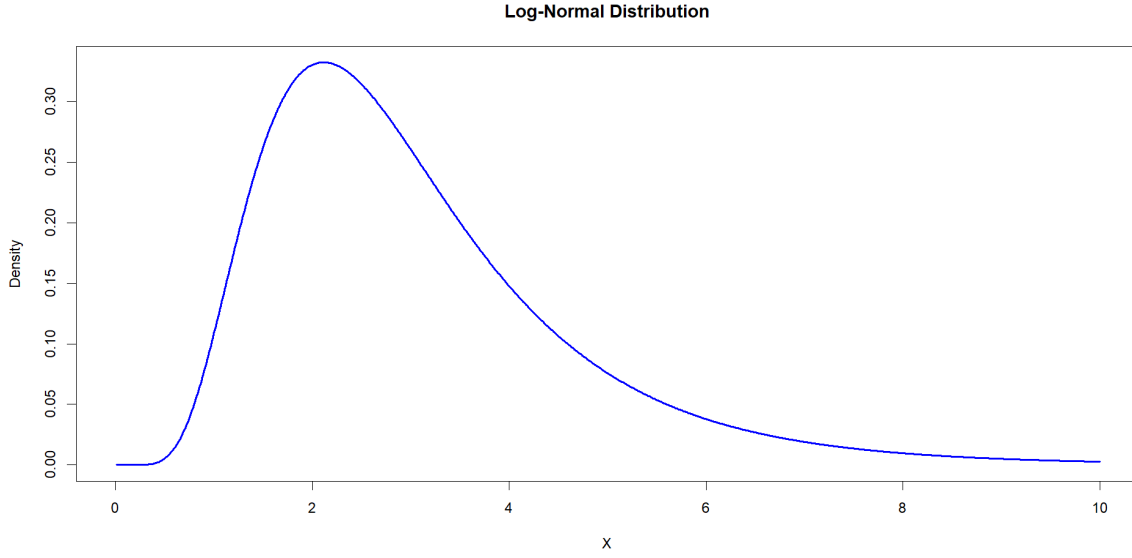
Fig. 3.2 Triangular distribution example (author's construction)



3.3.3 Log-normal distribution

The log-normal distribution is a probability distribution in which the natural logarithm of a random variable follows a normal distribution. The log-normal distribution differs from the normal distribution in several ways, with a major difference being its shape. While the normal distribution is symmetrical, the log-normal distribution is not. Due to the fact that the values in a log-normal distribution are positive, they create a right-skewed curve, as seen in the figure below (Zucchi, 2021). The log-normal distribution is commonly used in finance, particularly in option pricing via the Black-Scholes option pricing model or in models for changes in stock prices.

Fig. 3.3 Log-normal distribution example (author's construction)



The PDF for log-normal distribution can be mathematically explained as follows:

$$f(x) = \frac{1}{x\sqrt{2\pi\sigma_1^2}} e^{-\frac{(\ln(x)-\mu_1)^2}{2\sigma_1^2}}$$

Where:

$$\mu_1 = \ln\left(\frac{\mu^2}{\sqrt{\sigma^2 + \mu^2}}\right)$$

and

$$\sigma_1 = \sqrt{\ln\left(\frac{\sigma^2 + \mu^2}{\mu^2}\right)}$$

3.3.4 Uniform distribution

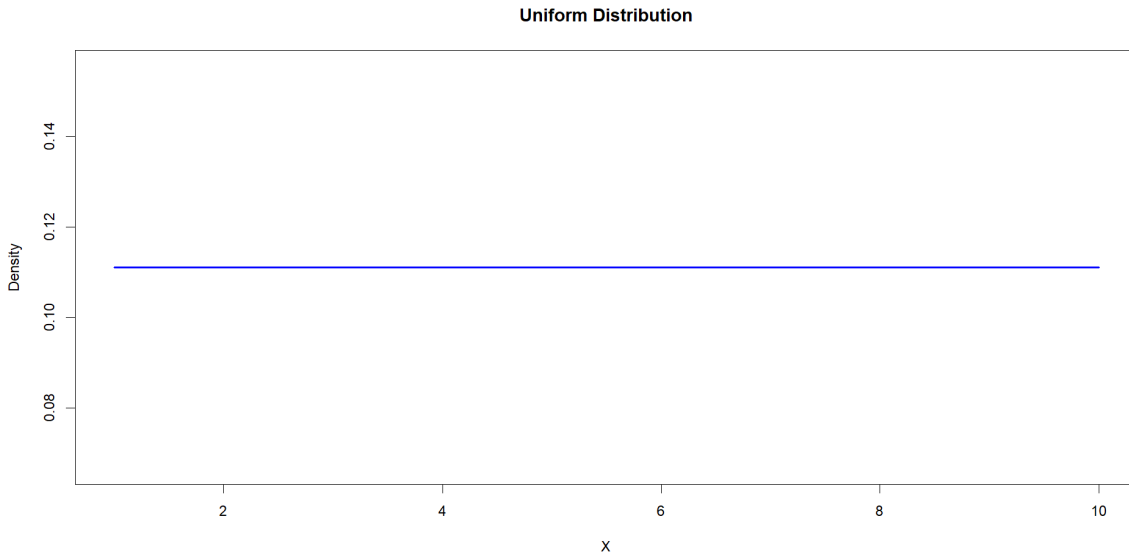
The uniform distribution is commonly used to model situations where all outcomes are equally likely, such as the roll of the dice. It is also commonly used in simulations and modelling problems, where a simple prior distribution is needed for unknown parameters. In finance, for instance, when predicting the possible value of free cash-flows, the uniform distribution can be used to model the range of likely interest rates, as all rates have the same likelihood of occurrence. A large number of iterations over this uniform distribution can then be generated by the Monte Carlo simulation, which helps in better estimating the future value of cash-flows. (Hayes, 2023)

Formally, a uniform distribution is defined by two parameters: the lower bound, "a", and the upper bound, "b". The probability density function (pdf) of a uniform distribution is represented by:

$$f(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b;$$

$$0 \text{ otherwise}$$

Fig. 3.4 Continuous uniform distribution example (author's construction)



As seen on the figure above, the probability of each number from 1 to 10 being chosen is the same (0.1), reflecting the uniform distribution of the data. However, values lower than 0 and larger than 10 have a zero probability of occurrence, indicating that the data is constrained within a certain range.

3.4 Kernel density estimation

Kernel density estimation is a statistical technique that is used for estimating the probability density function of a dataset. A smooth estimate of the density function is provided compared to traditional histograms, making it a useful tool for analysing probability distributions. All sample points are used in kernel estimation and it can suggest the presence of multiple modes in the distribution, unlike histograms. The effectiveness of kernel estimation relies on the shape of the kernel function and the coefficient of smoothness, which is a crucial parameter. (Weglarczyk, 2018) Monte Carlo technique can use Kernel density estimation to estimate the properties of the sampling distribution obtained from the simulations.

Simply put, Kernel is a non-parametric method to estimate the probability density function of random variable, which satisfies the three following properties:

a) It is symmetrical:

$$K(u) = K(-u)$$

b) The area under the curve of the function must be equal to one:

$$\int_{-\infty}^{\infty} K(u) du = 1$$

c) The value of Kernel function cannot be negative:

$$K(u) \geq 0 \text{ for all } -\infty \leq u \leq \infty$$

Since most of the distributions I work with in this thesis are either normal or log-normal, I can use simple Gaussian kernel function to estimate kernel density. (Pramanik, 2019) The Gaussian kernel function is explained by following equation:

$$K(x) = \frac{1}{h\sqrt{2\pi}} e^{-0.5(\frac{x-x_i}{h})^2}$$

“Where the value of x_i represents an observed data point, while x denotes the point at which the kernel function is being evaluated. The bandwidth parameter h determines the width of the kernel function.” (Pramanik, 2019)

The Kernel density estimator (KDE) is then computed simply by summing all the kernel values obtained by the function above. However, it still must be normalized by dividing the sum by the number of observations. (Pramanik, 2019)

$$KDE_j = \frac{1}{n} \sum_{i=1}^{i=n} \frac{1}{h\sqrt{2\pi}} e^{-0.5(\frac{x-x_i}{h})^2}$$

PRACTICAL PART

In this section of the thesis, a deeper examination of the concepts discussed in the previous section is undertaken, with a particular emphasis on practical applications. Specifically, the valuation of the business of the largest Czech car manufacturer, SKODA Auto, is carried out. A Monte Carlo simulation is then constructed, incorporating multiple variable inputs in order to obtain a more accurate estimate of the range within which the true value of the company lies.

First, the financials from the annual reports of SKODA for the past three years (2019 – 2021) were obtained. The actuals were inputted into the model to enable the creation of a forecast for the upcoming years. The financial statements for the next 5-year period were projected, and the Terminal value was calculated, which was discounted back to its Net Present value, representing the Enterprise value. In addition to the Enterprise value, a bridge was made to the Equity value to derive the value for shareholders only.

Secondly, the most crucial and variable inputs that were included in the model were identified. Then, a Monte Carlo simulation was executed with these inputs as variables, having a normal distribution and a given standard error. Upon completion, a normal distribution graph of the possible outcomes from the simulation was plotted.

The financial part of the work was conducted using Microsoft Excel, which is widely recognized as the best tool for financial analysis. In order to create a user-friendly financial model, I adhered to best practices for readability, such as incorporating colours, formatting, and links.

The econometric/statistical part of the thesis was carried out in R, a popular programming language for data analysis and statistical computing, which is widely used in the finance industry due to its extensive collection of modules for data analysis and its user-friendly syntax. To perform the analysis, packages such as dplyr, tidyr, and ggplot2 were utilized. With dplyr, it was easy to manipulate large datasets with minimal code, while tidyr was used to tidy data, i.e., restructure datasets to make them easier to work with. Finally, ggplot2 proved to be a versatile visualization package, allowing for the creation of a variety of graphs and plots to explore data.

4 FINANCIAL MODEL

4.1 Business description

The fundamentals and specifics of the business being valued are briefly described, starting with SKODA Auto, the largest Czech company (by revenues) and one of the biggest European car manufacturers, which is the subject of this thesis. SKODA Auto has a long history dating back to 1859, which makes it the fifth-oldest company producing cars. Over the years, SKODA Auto has become a crucial component of the Czech economy, with its production accounting for more than 5% of the Czech annual GDP in 2021. Additionally, SKODA Auto's manufacturing processes have created an entire supply chain, making the automotive industry the most important part of the Czech economy, accounting for more than 20% of the GDP. SKODA Auto sells its cars in over 100 countries worldwide, with annual sales commonly exceeding one million units.

Since 1991, SKODA Auto has undergone a gradual privatization process and has become a subsidiary of the Volkswagen Group, one of the three biggest car manufacturer concerns in the world. SKODA has been fully owned by the VW Group since 2000. As a result, despite its size, SKODA is not a publicly traded company, as Volkswagen is the sole shareholder. (SKODA, 2023)

4.2 Projecting the three financial statements

The first step in conducting a robust DCF analysis involves projecting the three main financial statements of the company. It is essential to forecast some of the P&L and cash-flow figures. Since the three statements are interconnected, the most effective approach to predicting those figures is to project all three statements and maintain their interdependence throughout the projection period. The last three years (2019 – 2021) serve as the benchmark for the upcoming years, and their historical data is taken into account during the forecasting process.

4.2.1 Profit & Loss Statement

The Profit & Loss Statement, also known as the Income Statement, is a crucial statement for the DCF analysis. The actual numbers from SKODA's annual reports were plugged into the left side of the statement. Then, as seen below the Net Income line, the main drivers of the past years were calculated. This step is crucial for identifying trends that will drive future performance. The main numbers on the P&L that drive the company's performance are typically:

- a) Revenue growth
- b) Gross margin

c) EBITDA margin

Apart from these, Operational Expenses as percentage of Revenue and Effective tax rate are also evaluated to further understand the historical effectiveness of the firm.

Tab. 4.1 Profit & Loss Statement projections (SKODA Annual reports, author's calculation)

Income Statement								
	2019A	2020A	2021A	2022P	2023P	2024P	2025P	2026P
Revenue	459 122	424 292	422 607	428 946	435 380	441 911	448 540	455 268
COGS	397 089	381 221	380 689	386 051	391 842	397 720	403 686	409 741
Gross Profit	62 033	43 071	41 918	42 895	43 538	44 191	44 854	45 527
SG&A expense	27 969	25 914	22 558	24 650	25 019	25 395	25 775	26 162
Other operating expense/(income)	(3 153)	(159)	(6 856)	(7 492)	(7 604)	(7 718)	(7 834)	(7 951)
OPEX	24 816	25 755	15 702	17 158	17 415	17 676	17 942	18 211
EBITDA	58 159	43 567	52 812	52 760	53 552	54 355	55 170	55 998
Depreciation & Amortization	20 942	26 251	26 596	27 024	27 429	27 840	28 258	28 682
EBIT	37 217	17 316	26 216	25 737	26 123	26 515	26 912	27 316
Net interest expense	681	2 295	1 356	1 700	1 700	1 700	1 700	1 700
Other financial expense/(income)	(1 959)	(2 842)	(2 460)	(1 700)	(1 700)	(1 700)	(1 700)	(1 700)
Pre-tax income	38 495	17 863	27 320	25 737	26 123	26 515	26 912	27 316
Taxes	6 809	2 688	4 910	4 890	4 963	5 038	5 113	5 190
Net Income to Company	31 686	15 175	22 410	20 847	21 159	21 477	21 799	22 126
<u>Drivers</u>								
Revenue growth		(7.6%)	(0.4%)					
Upside Case				5.0%	5.0%	5.0%	5.0%	5.0%
Base Case				1.5%	1.5%	1.5%	1.5%	1.5%
Downside Case				0.0%	—	—	—	—
Gross margin	13.5%	10.2%	9.9%	10.0%	10.0%	10.0%	10.0%	10.0%
OPEX as % of revenue	5.4%	6.1%	3.7%	4.0%	4.0%	4.0%	4.0%	4.0%
EBITDA margin	12.7%	10.3%	12.5%	11.5%	11.5%	11.5%	11.5%	11.5%
Effective tax rate	17.7%	15.0%	18.0%	19.0%	19.0%	19.0%	19.0%	19.0%

Based on the drivers from the past years, assumptions for the projection period are set. Depending on how fluctuant the actuals from the past years are, assumptions are either hardcoded (as in the model above) or extrapolated if there is a clear upside or downside trend. This is also the moment when equity research comes into play, as the assumptions that are plugged in have a tremendous effect on the outcome of the analysis (which will be subject to the Monte Carlo simulation later in the thesis).

For the Revenue growth, three possible scenarios which can be switched in the model were created. As the car manufacturing industry is currently undergoing a game-changing pivot towards electromobility, it seems harder to predict the growth than ever. Moreover, the current macroeconomic headwinds and supply chain shortages may cause further turmoil in this industry. (Burkacky et al., 2022) Because of this, and the fact that the car making industry is very mature (therefore not expected to grow at high digits), a rather conservative approach was taken with the growth scenarios. For the upside case, a 5% growth over the forecasted period was chosen. A 1.5% growth for the base case scenario seems to be around the consensus estimates of the big banks for the whole industry, and for the downside case, no growth whatsoever is predicted. The different scenarios and the respective probabilities of respective cases are explained in more detail in chapter 4.4.

Regarding the Gross Margin, a conservative estimate of 10% is assumed for all five years going forward. Similarly, the Operating Expenses as % of revenue and EBITDA Margin are also estimated based on a conservative approach. The assumption for the tax rate is that it will either remain constant at 19% or slightly increase. It is possible that the tax rate may

increase in the event of an upcoming recession where the government may need to boost state budget revenues. Therefore, for the projection, an increase in the effective tax rate is assumed due to the current macro environment (19% in the calculations).

When plugged into the model, most of the line items on the Income statement can finally be projected out, as seen above.

4.2.2 Balance sheet

Although the balance sheet itself is not always necessary for the DCF analysis, it is considered a best practice and serves as a sanity check for the financial model. Unlike the other two statements, the balance sheet is a snapshot of the company's finances. It provides an overview of what the company owns versus what it owes (i.e., its sources of funds and how it uses them). (Titman & Martin, 2016)

Tab. 4.2 Balance sheet & its projections (SKODA Annual reports, author's calculation)

	Balance Sheet							
	2019A	2020A	2021A	2022P	2023P	2024P	2025P	2026P
Cash and equivalents	45 753	18 669	22 422	23 884	25 191	26 335	27 308	28 103
Short-term financial assets	831	2 500	3 885	3 885	3 885	3 885	3 885	3 885
Accounts receivable, net	21 686	31 665	22 784	27 029	27 435	27 846	28 264	28 688
Inventories	24 863	24 516	31 901	29 696	30 142	30 594	31 053	31 519
Other current assets	6 978	5 982	5 569	6 000	6 000	6 000	6 000	6 000
Current Assets	100 111	83 332	86 561	90 495	92 653	94 660	96 510	98 195
Property, plant, and equipment, net	87 316	85 632	81 867	83 047	84 244	85 459	86 693	87 945
Identifiable intangible assets, net	39 422	44 600	48 697	48 900	49 105	49 314	49 526	49 741
Long term investments	2 950	2 956	3 162	3 000	3 000	3 000	3 000	3 000
Other long term assets	11 836	11 463	11 176	11 176	11 176	11 176	11 176	11 176
Long-Term Assets	141 524	144 651	144 902	146 122	147 525	148 949	150 395	151 862
Total assets	241 635	227 983	231 463	236 617	240 178	243 610	246 905	250 057
Accounts payable	70 267	66 153	58 230	63 461	64 412	65 379	66 359	67 355
Current provisions	19 217	26 269	29 078	29 078	29 078	29 078	29 078	29 078
Current Debt	1 752	1 376	1 720	1 720	1 720	1 720	1 720	1 720
Other current liabilities	15 903	12 836	14 811	14 517	14 055	14 461	14 344	14 286
Current Liabilities	107 139	106 634	103 839	108 775	109 265	110 637	111 501	112 439
Long-term debt	2 864	2 404	2 322	2 322	2 322	2 322	2 322	2 322
Provisions (Pension, Tax, Other)	15 169	17 242	16 757	16 757	16 757	16 757	16 757	16 757
Other non-current liabilities	6 837	6 783	7 017	7 017	7 017	7 017	7 017	7 017
Long-Term Liabilities	24 870	26 429	26 096	26 096	26 096	26 096	26 096	26 096
Total Liabilities	132 009	133 063	129 935	134 871	135 361	136 733	137 597	138 535
Stockholder's Equity	109 626	94 920	101 528	101 746	104 817	106 876	109 308	111 522
Total Liabilities + Stockholder's Equity	241 635	227 983	231 463	236 617	240 178	243 610	246 905	250 057
Check	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

As with the Income statement, the numbers from the past three years have been inputted. However, if done correctly, most of the items on the Balance sheet are connected to other statements or have their own schedules.

Tab. 4.3 Working capital supporting schedule (SKODA Annual reports, author's calculation)

Working Capital Schedule								
	2019A	2020A	2021A	2022P	2023P	2024P	2025P	2026P
<u>Working Capital Balances</u>								
Accounts receivable, net	21 686	31 665	22 784	27 029	27 435	27 846	28 264	28 688
Inventories	24 863	24 516	31 901	29 696	30 142	30 594	31 053	31 519
Total non-cash current assets	46 549	56 181	54 685	56 726	57 577	58 440	59 317	60 207
Accounts payable	70 267	66 153	58 230	63 461	64 412	65 379	66 359	67 355
Total non-debt current liabilities	70 267	66 153	58 230	63 461	64 412	65 379	66 359	67 355
Net working capital	(23 718)	(9 972)	(3 545)	(6 735)	(6 836)	(6 938)	(7 042)	(7 148)
Change in net working capital		13 746	6 427	(3 190)	(101)	(103)	(104)	(106)
<u>Drivers</u>								
Days sales outstanding (DSO)	17	27	20	23	23	23	23	23
Inventory Turns	16.0x	15.5x	11.9x	13.0x	13.0x	13.0x	13.0x	13.0x
Days payables outstanding (DPO)	65	63	56	60	60	60	60	60

In the model, two extra schedules were created to further decompose the items of this statement. The first schedule focuses on the short-term side of the balance sheet and is called the working capital schedule. The main driver of working capital are:

- Days sales outstanding (DSO)
- Inventory turns
- Days payable outstanding (DPO)

As can be seen in the schedule, the fluctuation of SKODA's DSO between 17 and 27 in the years leading up to the forecast is noted. For the purpose of projection, a DSO of 23 is calculated, and a slightly higher DSO is expected in the upcoming years due to the potential recession, making it slightly more difficult for SKODA to collect its revenues.

Days payable outstanding, on the other hand, indicates how many days it takes the company to pay down its debts. Basically, it is the average time for the company to pay its bills. For the projection period, DPO is assumed to remain at 60, based on the past performance as a reference.

Inventory turns measure how often inventory is replaced relative to a company's revenues. Here, an estimate of 13x inventory turns has been determined based on the average of past years. This indicates that the company replaces its inventory thirteen times throughout the financial year.

When the main drivers are identified and estimated, the current assets and current liabilities are used to calculate NWC for the upcoming years. Then, NWC is calculated as the difference between the current assets and current liabilities in the respective year. The single most important figure of this Working capital schedule is the Change in NWC (because it directly plugs into the cash flow statement). In the projection years, the annual change in working capital is almost negligible. This is partly because the change in NWC is not meant to significantly affect the model. (Wall Street Oasis, 2023) (Darmodaran, 1994) (Viernimmen et. Al, 2022)

Tab. 4.4 PP&E and Intangible supporting schedule (SKODA Annual reports, author's calculation)

PP&E and Intangible Schedule								
	2019A	2020A	2021A	2022P	2023P	2024P	2025P	2026P
Beginning PP&E Balance			85 632	81 867	83 047	84 244	85 459	86 693
Depreciation			(19 947)	(20 268)	(20 572)	(20 880)	(21 193)	(21 511)
PP&E CapEx			16 182	21 447	21 769	22 096	22 427	22 763
Ending PP&E			81 867	83 047	84 244	85 459	86 693	87 945
Beginning Intangible Balance			44 600	48 697	48 900	49 105	49 314	49 526
Amortization			(6 649)	(6 756)	(6 857)	(6 960)	(7 064)	(7 170)
Additions			10 746	6 959	7 063	7 169	7 276	7 386
Ending Intangible Balance			48 697	48 900	49 105	49 314	49 526	49 741
<u>Drivers</u>								
Capex as % of revenue			3.8%	5.0%	5.0%	5.0%	5.0%	5.0%
D&A as % of revenue			6.3%	6.3%	6.3%	6.3%	6.3%	6.3%

In the PP&E and Intangible schedule, assumptions are made regarding the further development of non-current tangible and intangible assets. The difference between the ending and beginning balances of PP&E and intangibles is determined by new investments and depreciation and amortization, respectively. CapEx and D&A are key drivers and are predicted to be a stable percentage of revenue for the given year. Additionally, it is assumed that amortization accounts for 25% of the D&A as a whole, and the additions to intangibles grow by 3% annually.

Overall, it is not expected that there will be any significant changes on the assets, liabilities, and equity sides of the balance sheet. This is due to two main reasons. Firstly, the debt has become more expensive as a result of rising interest rates. Secondly, SKODA is a mature company, and therefore it is assumed that the retained earnings would not be as volatile as they might be for a growing company.

4.2.3 Cash flow statement

The cash flow statement is a financial statement that summarizes the movement of cash and cash equivalents that come in and go out of a company. Technically speaking, the cash flow statement itself is not crucial for the DCF valuation as most of the inputs come from the Income statement and we are looking for unlevered free cash flow (which cannot be found in the conventional cash flow statement). However, just like the balance sheet, it helps us to better understand the company, and it is always good practice to project all the statements. Furthermore, the ending balance in cash flows directly affects the balance sheet, and therefore, it is important for the successful completion of the balance sheet. (Murphy, 2023)

The conventional cash flow is divided into three segments:

- a) Operational cash flow
- b) Investing cash flow
- c) Financing cash flow

The operational cash flow shows the ability of the business to generate cash from its day-to-day operations. As can be seen in the table below, the process begins by adding back depreciation and amortization to the net income because D&A are non-cash expenses. Since car manufacturing is an asset-heavy industry, the add-backs are key components of the cash

flow calculation for SKODA. In addition, we add or subtract the change in NWC (calculated above). Based on the conservative assumptions, I expect only very slight growth in cash flow from operations throughout the years.

Investing cash flow summarizes the investing activities that involve the acquisition or disposal of long-term assets and investments, such as capital expenditures and capitalization of development costs, that use or generate cash. In the case of SKODA, these activities typically involve expenses that use cash.

Financing cash flow usually comprises debt issuance or stock repurchase (in case of publicly traded companies). (Viernimmen et. al, 2022) As we can see below, SKODA dividend payments to shareholders are the major cash outflow in this area. It is quite difficult to predict the dividend pay-out ratio and so based on the previous years I assume this ratio to be around 90 %. Because of the forementioned fact that it is hard to predict dividend payout (especially when the firm is not public), the annual dividend was kept (relatively) at the similar amount throughout the projection.

Tab. 4.5 Equity schedule (SKODA Annual reports, author's calculation)

Equity Schedule								
	2019A	2020A	2021A	2022P	2023P	2024P	2025P	2026P
<u>Stockholder's Equity</u>								
Beginning balance				101 528	103 613	105 729	107 876	110 056
Net income				20 847	21 159	21 477	21 799	22 126
Dividends				(18 762)	(19 044)	(19 329)	(19 619)	(19 913)
Ending balance				103 613	105 729	107 876	110 056	112 269
<u>Dividends</u>								
Dividend payout ratio	91.2%	208.8%	67.7%	90.0%	90.0%	90.0%	90.0%	90.0%
Dividends	28 890	31 680	15 170	18 762	19 044	19 329	19 619	19 913

Predicting future leverage for SKODA is challenging, but it is possible that the company may benefit from cash-pooling, which is a method of allocating cash between companies within a group. As SKODA is part of the VW group, this could mean that there may not be a significant need to raise debt levels in the future.

Tab. 4.6 The cash flow statement projection (SKODA Annual reports, author's calculation)

Cash Flow Statement								
	2019A	2020A	2021A	2022P	2023P	2024P	2025P	2026P
Net Income				20 847	21 159	21 477	21 799	22 126
Depreciation				20 268	20 572	20 880	21 193	21 511
Amortization				6 756	6 857	6 960	7 064	7 170
(Increase) Decrease to Net Working Capital				6 434	6 531	6 629	6 728	6 829
Cash Flow from Operations				54 305	55 119	55 946	56 785	57 637
CapEx			(21 136)	(21 453)	(21 775)	(22 101)	(22 433)	(22 769)
Additions to capitalized dev. costs			(12 271)	(12 627)	(12 994)	(13 371)	(13 760)	(14 159)
Cash Flow From Investing Activities				(34 080)	(34 769)	(35 473)	(36 192)	(36 929)
Dividends				(18 762)	(19 044)	(19 329)	(19 619)	(19 913)
Cash Flow From Financing Activities				(18 762)	(19 044)	(19 329)	(19 619)	(19 913)
Beginning Cash				22 422	23 884	25 191	26 335	27 308
Change in Cash				1 462	1 307	1 144	974	795
Ending Cash				23 884	25 191	26 335	27 308	28 103

After projecting all three financial statements, it can be observed that the annual change in cash remains relatively constant throughout the projection period. As a result, the DCF analysis can now be performed.

5 Deterministic DCF analysis

5.1 WACC calculation

When assessing the DCF analysis, one of the key questions is to determine the average cost of capital, WACC. The actual cost of capital used for future cash-flow discounting is demonstrated below, showing how it was derived.

Tab. 5.1 Weighted Average Cost of Capital calculation (SKODA Annual reports, author's calculation)

Weighted Average Cost of Capital (WACC)	
<u>Basic Capitalization Table:</u>	
Short-term debt	1 720
Long-term debt	2 322
Total Debt	4 042
 Total Equity Value	 101 528
Total Capitalization	105 570
Debt as % of Total Capitalization	3.8%
Equity as % of Total Capitalization	96.2%
Current Levered Beta (per Capital IQ)	1.47
<u>Target Capitalization Ratios:</u>	
Debt as % of Total Capitalization	3.0%
Equity as % of Total Capitalization	97.0%
<u>Cost of Debt:</u>	
Debt Financing Interest Rate	5.0%
Corporate Tax Rate (Debt Shield)	19.0%
Tax-affected Cost of Debt	4.1%
<u>Cost of Equity:</u>	
Risk-free Rate	0.1%
Equity Risk Premium	7.5%
Size "Premium"	1.2%
Levered Beta	1.47
Cost of Equity	9.9%
Weighted Avg. Cost of Capital (WACC)	9.7%

The capital structure, beta, return on equity (ROE) and cost of debt are key parameters that need to be known in order to correctly estimate WACC. Total capitalization, which is the sum of total debt and total equity value, can be easily obtained from the prepared balance sheet discussed in section 4.2.2. It can be observed that SKODA does not have a significant amount of debt. Therefore, it can be suggested that the main driver of WACC will be the cost of equity. The levered beta has been obtained from Capital IQ, a knowledge base that provides users with detailed information on not only publicly traded companies. As SKODA is part of the Volkswagen Group, the beta of the group is being used. (Murphy 2, 2023)

An information on cost of debt is usually not easily found. It is influenced by many factors such as the current interest rates, the riskiness of the company, and others. Some estimates were found in the annual reports, however, the true cost of debt varies with every bond issuance or new loan. Fortunately, as mentioned earlier, debt represents only a small portion of total capitalization, therefore, an assumption of 5% debt cost has been made."

The cost of equity is easier to estimate (at least for big, mature companies such as SKODA/Volkswagen Group) since CAPM can be used, as explained in detail in section 2.3.1. When the cost of equity is calculated using CAPM, a cost of 9.9% is arrived at. At first, this number seemed a bit high as it was expected to be around 6-7%, but considering factors such as high inflation, supply chain issues, or the switch to electromobility, it may be justified.

Overall, we get a WACC of 9.7 %, which is nothing more than just the average of cost of debt and cost of equity we calculated.

Taking advantage of the comparable company's Betas can also be done in addition to calculating a company's Beta in relation to the market. However, to account for varying capital structures, each Beta can be first "unlevered," then the average can be taken, and finally the Beta can be "re-levered" using the target's (or average) capital structure. This approach can be useful when attempting to assess the Beta of a private company or when verifying the result of a calculation. (Pignataro, 2013)

Although the beta of Volkswagen Group was obtained from Capital IQ, benchmarking it with comparable publicly traded automobile manufacturers was also conducted. This involved obtaining information on capitalization structure and levered beta, using public resources or annual reports, and then unleveraging the betas. Afterward, using the average debt/equity structure of the peer group and the company's known tax rate, the median beta of the group was leveraged to estimate the company's beta. Based on the selected companies, the calculated beta was slightly lower than the beta based on VW CAPM.

Tab. 5.2 Beta comparable comparisons (Annual reports & Capital IQ, author's calculation)

Beta Comparable Comparisons:							
	Equity				Market	Corporate	Unlevered
	Debt Out.	Market Cap	Total Cap.	Debt / Equity	Lev. Beta	Tax Rate	Beta
Bayerische Motoren Werke AG	105 968	47 485	153 453	223.2%	1.18	19.0%	0.42
Mercedes-Benz Group AG	116 021	59 097	175 118	196.3%	1.39	19.0%	0.54
Volvo Group	€18 874	€5 544	24 418	340.4%	1.18	20.6%	0.32
Renault SA	60 185	7 700	67 885	781.6%	1.85	26.5%	0.27
General Motors Company	€110 985	€55 877	166 862	198.6%	1.25	21.0%	0.49
Ford Motor Company	€128 932	€59 820	188 751	215.5%	1.29	21.0%	0.48
Median	108 476	51 681	160 157	209.9%	1.27	20.8%	0.45
Comp Set Median Unlevered Beta	0.45						
Debt / Equity	209.9%						
Marginal Tax Rate	19.0%						
Levered Beta	1.21						

5.2 Unlevered free cash flow calculation

Once the cost of capital is known, it can be utilized to discount the future cash flows, accounting for the time value of money. The total revenues are projected to grow by 1.5% annually. Monte Carlo simulation is a good use case for this scenario, as I elaborate later in

the thesis. Since the business's growth rate cannot be accurately determined, it is easier to estimate a possible range of growth and run a random simulation multiple times. This logic applies to most of the figures mentioned in this section.

The assumption is made that the gross margin will remain the same. This is due to the predictability of the car manufacturing business, where margins tend to remain relatively stable. All of the line items used for projecting future cash flows come either from the income statement or the balance sheet supporting schedules, which are explained in detail in sections 4.2.1 and 4.2.2. The process of deriving unlevered free cash flow is elaborated on in chapter 2.1.

When valuing cash flows, two methods are used to determine the period: the mid-year convention and the end-of-year convention. With the end-of-year convention, each cash flow is discounted at the end of a full year. Year 1 is discounted at the end of the first year, year 2 at the end of the second year, and so on. With the mid-year convention, each cash flow is discounted by half a year. Year 1 is discounted by half a year (0.5), year 2 by 1.5, and so on. Since it is assumed that the cash flows are generated evenly throughout the year, using the mid-year convention is the preferable variant. Additionally, SKODA is not a seasonal business, further supporting this concept. (Pignataro, 2013)

Tab. 5.3 NPV of Unlevered free cash flows (Annual report, author's calculation)

Unlevered Free Cash Flow						
	FY 2021	FY 2022	FY 2023	FY 2024	FY 2025	FY 2026
Total Revenues	422 607	428 946	435 380	441 911	448 540	455 268
Growth Rate (%)		1.5%	1.5%	1.5%	1.5%	1.5%
Cost of Sales	380 689	386 051	391 842	397 720	403 686	409 741
Gross Profit	41 918	42 895	43 538	44 191	44 854	45 527
Gross Margin (%)	9.9%	10.0%	10.0%	10.0%	10.0%	10.0%
SG&A expense	22 558	24 650	25 019	25 395	25 775	26 162
Other expense/(income)	(6 856)	(7 492)	(7 604)	(7 718)	(7 834)	(7 951)
Total SG&A	15 702	17 158	17 415	17 676	17 942	18 211
EBITDA	52 812	52 760	53 552	54 355	55 170	55 998
EBITDA Margin (%)	12.5%	12.3%	12.3%	12.3%	12.3%	12.3%
Less: Depreciation & Amortization	26 596	27 024	27 429	27 840	28 258	28 682
EBIT	26 216	25 737	26 123	26 515	26 912	27 316
Corporate Tax Rate	19.0%	19.0%	19.0%	19.0%	19.0%	19.0%
Tax-affected EBIT (NOPAT)	21 235	20 847	21 159	21 477	21 799	22 126
Add-back: D&A (non-cash expense)	26 596	27 024	27 429	27 840	28 258	28 682
Tax-affected EBIT + D&A	47 831	47 870	48 588	49 317	50 057	50 808
Less: CapEx	(21 136)	(21 453)	(21 775)	(22 101)	(22 433)	(22 769)
Plus / (Less): Change in Net Working Capital (NWC)	(6 427)	3 190	101	103	104	106
Unlevered Free Cash Flow (FCF)	20 268	29 607	26 915	27 318	27 728	28 144
Mid-Year Convention?	Yes					
Discount Year	0.0	0.5	1.5	2.5	3.5	4.5
Discount Factor		0.95	0.87	0.79	0.72	0.66
NPV of Unlevered FCF		28 262	23 409	21 644	20 018	18 513
NPV of FY22 - FY26 Free Cash Flow		111 846				

5.3 Terminal value calculation

In this section, the approaches for assessing the terminal value estimation are explained. The perpetuity method is used in this paper, which involves a steady stream of cash flows with no end. After obtaining the terminal value, it is added to the sum of the present value

of the company's projected cash flows to derive the total enterprise value for the business. (Pignataro, 2013)

Some information regarding the perpetual growth rate has been found in the 2021 annual report (page 251) of Volkswagen Group. It states: *"The estimates for the cash flows following the end of the planning period are generally based on a growth rate of up to 1% p.a."* Since a conservative approach is preferred in this analysis, a perpetual growth rate of 0.5% was chosen, which appears to be achievable in the long-term.

It can be observed that despite the perpetual growth rate being only 0.5%, the terminal value of the business is a significant driver of the enterprise value. Additionally, the EV multiple implied in 2026 is slightly lower than the current one. A very low EV/EBITDA multiple of 5.8 is calculated, and a valuation of around 10x EV/EBITDA would be expected for the car manufacturing business. (Mascarello, 2013) However, private companies usually sell for less, mainly because they are less liquid.

The enterprise value for the neutral case is calculated to be **CZK 313 billion**. If the downside case is chosen, the enterprise would be valued at around 294 billion, whereas the upside case would result in an enterprise value of 334 billion.

Tab. 5.4 NPV of Terminal value (Annual report, author's calculation)

Terminal Value Using Perpetuity Approach to Present Value			
	FY 2026	Terminal Yr. FY 2027	Perpetual Growth Rate
Total Revenues	455 268	457 544	0.5%
Cost of Sales	409 741	411 790	
Gross Profit	45 527	45 754	
Gross Margin (%)	10.0%	10.0%	
SG&A expense	26 162	26 293	0.5%
Other expense/(income)	(7 951)	(7 991)	0.5%
Total SG&A	18 211	18 302	
EBITDA	55 998	56 278	
EBITDA Margin (%)	12.3%	12.3%	
Less: Depreciation & Amortization	28 682	28 825	0.5%
EBIT	27 316	27 453	
Corporate Tax Rate	19.0%	19.0%	
Tax-affected EBIT (NOPAT)	22 126	22 237	
Add-back: D&A (non-cash expense)	28 682	28 825	
Tax-affected EBIT + D&A	50 808	51 062	
Less: CapEx	(22 769)	(22 883)	0.5%
Plus / (Less): Change in Net Working Capital (NWC)	106	106	
Unlevered FCF	28 144	28 284	
Discount Year	4.5	5.5	
Discount Factor	0.66		
Terminal Value in FY26		305 817	
PV of Terminal Value today		201 165	
NPV of Terminal Value	201 165		
		% of TEV	
NPV of FY22 - FY26 Free Cash Flow	\$111 846	35.7%	
NPV of Terminal Value	\$201 165	64.3%	*Significant value driven by terminal value of business
Total Enterprise Value Using DCF	313 011	100.0%	
Multiple on FY2022 EBITDA	5.9x		
Multiple on FY2026 EBITDA	5.6x		

The advantages of incorporating Monte Carlo simulations into the DCF model will be illustrated in the upcoming chapter, showing how this method can provide more robust estimates for the enterprise value compared to the standard discounted cash flow (DCF) method.

6 Monte Carlo incorporation to the DCF

In the Monte Carlo version of the DCF model, the initial assumptions remain unchanged, but the assessment of those assumptions differs. Instead of using point estimates in the deterministic DCF model, specific probability distributions are used for each parameter. This approach enables the full utilization of the central limit theorem. As a result, the more simulations that are run, the closer the estimated enterprise value is to the actual value. Additionally, Monte Carlo simulations provide the sampling distribution of the entire range of possible values for the company, offering greater confidence in the results of the DCF. (Ali et al., 2010) (Edner & Paulsson, 2013) (French & Gabrielli, 2005) (Kenton, 2023)

6.1 Input variables discussion

To begin with, the behaviour of different input variables of the model needs to be defined. It is likely that each variable will develop in a different manner since they are affected by various circumstances.

6.1.1 Revenue growth

In the car manufacturing industry, multiple factors such as consumer demand, regulations, competition, or new products can affect revenue growth rates. Expecting a normal distribution of revenue growth rates might be a bit unrealistic, however, it is probably the most convenient estimate we can make. The industry may experience periods of high growth when the economy is booming, and new models or technologies are being introduced. On the other hand, growth rates may slow down during an economic downturn, as seen in 2008 and most recently in 2020.

6.1.2 EBITDA Margin

It is suggested that the distribution of EBITDA margins might be relatively stable due to the mature nature of the car manufacturing industry. The assumption is that the EBITDA margins will not fluctuate significantly over the projection period. Additionally, since the EBITDA margin is calculated by dividing EBITDA by Sales, it is unlikely that the margin will turn negative, even if sales decline. It is possible that the distribution of EBITDA margins in the industry may be positively skewed, given the assumption that there are more low growth rates and fewer high growth rates in mature industries such as car manufacturing. However, the degree of skewness is uncertain.

A log-normal distribution is commonly used to model such cases, where there is no negative values and a positive skewness. Assuming that the logarithm of the growth rate is normally distributed, I think that using log-normal distribution could serve as useful estimation of the actual distribution of such unpredictable variable.

6.1.3 Operational expenses

The distribution of operational expenses as a percentage of revenue in car manufacturing can be influenced by various factors, such as the size of the company, the type of vehicles produced, the location of manufacturing facilities, and the level of automation in production processes. (Paoli et. al, 2022)

That being said, it is often observed that the distribution of operational expenses as a percentage of revenue in car manufacturing is positively skewed, with a long tail on the right side of the distribution. This is because high fixed costs associated with operating a manufacturing plant, such as labour, equipment, and facility costs, can be spread out over a large volume of production. The distribution may also be influenced by various factors such as the size of the company, the type of vehicles produced, the location of manufacturing facilities, and the level of automation in production processes.

Moreover, since operational expenses cannot be negative, we can also consider using a log-normal distribution. However, given that margins in the industry are generally stable, the standard deviation is likely to be a small number.

6.1.4 Capital Expenditures

The log-normal distribution is again considered to be the best probability distribution that describes CapEx as a percentage of revenue in the model, due to the assumption that it is skewed to the right. It is known that the variable is always positive, and there are usually no significant differences year on year. However, if a substantial investment in infrastructure or equipment is required by SKODA for competitive or regulatory reasons, this could result in a spike in capital expenditures in some years. Alternatively, SKODA may need to scale back spending due to macroeconomic headwinds, such as high inflation and interest rates, resulting in lower-than-usual CapEx.

In the current environment, it is less likely that CapEx will exceed the average spending seen in previous financial statements compared to a potential downswing in spending. Therefore, I strongly believe that all these factors contribute to a long-tailed log-normal distribution of CapEx as a percentage of revenue.

6.1.5 Net Working Capital

Since Net Working Capital can be both positive and negative, the log-normal distribution is not a suitable choice. Additionally, it is typically affected by small factors that tend to even out in the long-term, and it is challenging to identify any patterns in these factors (e.g., supplier/customer behavior, overall conditions, etc.). Therefore, a normal distribution may be the best fit to account for the randomness of Net Working Capital.

6.1.6 WACC

It is not expected that WACC would be skewed towards any side of the axis, and therefore, the best estimate would follow the normal distribution pattern. The Central Limit Theorem

can be a useful concept to think about, as if the sample is sufficiently large (which it is), the mean of a random sample will be normally distributed. Additionally, since the WACC was calculated thoroughly, there is no desire for it to be substantially different in various iterations.

6.1.7 Terminal Growth

The terminal growth is usually not significantly dependent on other variables in the model, which makes it a suitable candidate for a uniform distribution. Additionally, due to a lack of prior data on terminal growth, in contrast to financials, it may be appropriate to estimate it using a uniform distribution. Furthermore, we are limited by inflation or GDP growth rate when it comes to terminal growth. Uniform distribution is useful in this regard because it is bounded, assigning equal probability to all possible values within the specified range. (Damodaran, 1994)

The complete list of distributions used in the model, including mean values and standard deviations are disclosed in the table below.

Tab. 6.1 Probability distributions of key DCF drivers – Base case (own assumptions)

DCF Drivers	Probability distribution	Mean value	Standard deviation
Revenue Growth	Normal	0.015	0.010
EBITDA Margin	Normal	0.132	0.010
D&A as % of Revenue	Log-Normal	0.063	0.100
CapEx as % of Revenue	Log-Normal	0.050	0.300
NWC as % of Revenue	Normal	(0.016)	0.010
WACC	Normal	0.098	0.010
Terminal Growth	Uniform	0.005	~ 0.001

It is important to note that the distributions shown in the table above relate only to the Base case. The other cases will be described on the following pages. However, for the purpose of simplification, the only variable that changes is the revenue growth, while the other variables are held constant in all the cases yet to be discussed.

6.2 Introduction of different growth scenarios

Now that probability distributions have been assigned to the main drivers of company value, different growth scenarios should be explored. To keep things brief, only the topline growth will be analyzed in depth. While margins and other variables could be adjusted in different scenarios, doing so would introduce unwanted complexity to the model. Therefore, only revenue growth scenarios will be examined and how they are impacted under various circumstances.

As already briefly mentioned in chapter 4.2.1, three main scenarios will be considered. However, since predicting and planning for upsides of the business might be easier, downside risks are usually not as clear and predictable. Due to this, a so-called "black swan" scenario has been introduced, which covers unforeseen events that the industry may face in the coming years. The section below provides more details on each of the topline scenarios for SKODA Auto.

6.2.1 Upside case

In June 2021, the Company introduced a new development plan for the third decade titled NEXT LEVEL – SKODA STRATEGY 2030. This aims to build on the success of Strategy 2025+ to accelerate in the sphere of internationalization, electrification, and digitization. (SKODA, 2021).

SKODA Auto Group delivered 731,300 vehicles worldwide in 2022 and achieved a solid financial result despite a highly challenging market environment. It increased its revenues by 18.5% compared to the previous year (even though this comparison is a bit tricky due to Covid pandemic in 2021). Although the ongoing semiconductor shortage, supply chain issues, and increase in raw material prices had a significant impact on the financial result, SKODA Auto managed to achieve a positive operating profit of CZK 15 billion. The company's financial strength and resilience give a promising outlook for the upcoming years. With the Next Level Efficiency+ program implemented and running on track, SKODA Auto plans to invest heavily in electric mobility and digitalization, with €134 billion allocated for e-mobility and CZK 17 billion for digitalization by 2027. The company is also pushing internationalization and targeting new growth markets and regions like South-East Asia and India.

Considering SKODA Auto's financial strength and investments in future technologies, the company is poised for further growth in the upcoming years. If the market environment remains stable and everything goes as planned, SKODA Auto can achieve a 5% YoY revenue growth in the next five years. The company's focus on electric mobility and digitalization is in line with the industry's trend, and its investments in these areas should help the company maintain its position in the competitive automotive market. (SKODA, 2023)

6.2.2 Base case

According to McKinsey's report, the automotive software and electronics market worldwide is estimated to grow at a compound annual rate of 5.5 percent from 2019 to 2030. On the other hand, the overall automotive market for passenger cars and light commercial vehicles (LCVs) is predicted to have a slower growth rate of 1 percent during the same period, with an increase from 89 million units in 2019 to 102 million units in 2030. (Burkacky et al., 2023) (Von Paul et al., 2023)

It is assumed that SKODA Auto's investment in e-mobility and digitalization could help the company to slightly outperform this overall market growth rate. Therefore, the combination of SKODA Auto's investment in electric mobility and digitalization, as well as the predicted growth in the automotive software and electronics market, provide a strong basis for

assuming that the company can achieve an annual growth rate of 1.5% in the upcoming 5 years.

6.2.3 Downside case

SKODA Auto faces several potential factors that could lead to a period of no revenue growth in the upcoming years. To name a few, the challenges include global recession, supply chain issues, regulatory changes, or new competitor on the market.

To name a few potential factors that could push the global economy into recession, firstly, even a moderate negative shock given the current weak world economy growth outlook could be detrimental. Historically, all global recessions since 1970 were preceded by a year of relatively weak global growth. Indicators such as world industrial production, trade, and oil consumption have tended to slow down in the two years before global recessions, and several high-frequency indicators of activity have weakened over the past year. Secondly, the recent slowdown in global GDP growth is mainly due to pronounced declines in growth in several major economies, including the United States, which contracted by an annualized rate of 0.9 percent in the second quarter of 2022. All previous global recessions coincided with sharp slowdowns or outright recessions in several major economies. Amid a substantial increase in the cost of borrowing, a sharp slowdown in the United States and other major economies could trigger acute financial stress, resulting in a significant deterioration in global growth. (Guenette et al., 2022)

During the last worldwide financial crisis, between the years 2008 and 2009, SKODA Auto experienced a decrease in revenue due to the crisis and its impact on the automotive industry. As reported in SKODA Auto's annual reports, the company's revenue was approximately CZK 92.4 billion in 2008, while in 2009, the revenue decreased to approximately CZK 77.3 billion, representing a year-on-year decline of approximately 16.3%. This decline was attributed to lower sales volumes, reduced demand, and production shutdowns during the recession. However, SKODA Auto was able to weather the storm and remained profitable during this period, and the company's revenue growth recovered strongly in the years that followed. Since it is very difficult to predict how the next global crisis might unfold, a rather conservative approach is taken and it is not expected that SKODA will recover as well in the case of such a recession.

Since the globalized world is quite interconnected these days, recession is usually expected to be caused by multiple correlated events. As seen during Covid-19 pandemic, the supply chain can be quite a problem for manufacturing businesses. SKODA Auto, like other automotive manufacturers, has faced supply chain bottlenecks at times, and here are some specific examples (SKODA, 2023):

- In 2018, SKODA Auto was forced to halt production at its Czech Republic factories for several days due to a shortage of engines from its supplier, Volkswagen Group. The bottleneck was caused by a fire at a parts supplier in Germany, which disrupted production of critical engine components. As a result, SKODA Auto's revenue growth slowed from 20.8% in 2017 to 4.4% in 2018.

- In 2020, SKODA Auto, along with many other automotive manufacturers, faced supply chain disruptions due to the COVID-19 pandemic. The pandemic caused factory shutdowns and production slowdowns across the world, leading to shortages of critical components such as semiconductors and microchips. As a result, SKODA Auto and other manufacturers were forced to adjust production schedules and reduce output. The COVID-19 pandemic and associated supply chain disruptions caused SKODA Auto's revenue to decline by 13.9% year-on-year compared to 2019.

In general, SKODA faces several potential issues that could go wrong, but most of these are closely tied to a global economic downturn. Therefore, they would be viewed more as a consequence of a recession rather than an independent occurrence. To name a few potential byproducts of a recession, we can mention sharp fall in consumer demand which in turn results in lower revenue. (KPMG, 2022)

6.2.4 Worst case

Speaking about the “black swan” events which have nonnegligible probability of occurrence, we can think for instance of another global pandemic outbreak, war conflict or some kind of severe political unrest. These events have potential to halt SKODA’s production capabilities for the sometime. Probability of such incidents are almost impossible to predict, however the current environment has shown us that such events can happen and can have severe consequences, like in the case of war in Ukraine, which resulted in sharp rise in the price of some crucial commodities. Even though European countries were successful in mitigating the worst case by subsidizing the business and people, the shadow of anything worse happening in the future prevails (Guenette et al., 2022).

Tab. 6.2 Topline growth scenarios (World Bank, own assumptions)

Scenario	Topline Growth	Scenario Probability
Upside case	5.0%	10%
Base case	1.5%	70%
Downside case	0.0%	15%
Worst case	(10.0%)	5%

Based on the trends and factors presented in the section above, the following probabilities and growth rates were assigned to each scenario: Upside case (10% probability, 5.0% growth) reflects the potential positive impact of favourable market conditions and successful implementation of growth strategies; Base case (70% probability, 1.5% growth) represents the most likely scenario, taking into account the current economic situation and industry trends; Downside case (15% probability, zero growth) reflects the possibility of unfavourable market conditions and operational challenges; and Worst case (5% probability, negative 10.0% growth) represents a potential extreme scenario, such as a significant economic downturn, major industry disruption or any other unforeseen event.

6.3 Simulating the Model

In this section, it is explained how the set up for Monte Carlo simulation has been done.

In the beginning, the free cash flow of SKODA is projected over a 5-year period. This is calculated in the same way as in the Excel model. The years of interest are defined and the last known revenue figure is included. Then, all the parameters used for the single value estimate are hardcoded, which will later be used in the program as mean values of specific probability distributions. The parameters include the sales growth rate, EBITDA margin, D&A as a percentage of revenue, NWC as a percentage of revenue, and a few more parameters.

Using a for-loop, the code then calculates revenues in the projected years, based on the given growth rates.

Once the main variables have been defined, the static part of the analysis in R can be completed by inputting the WACC and Terminal growth rate. While Excel provides more comfort in working with single value estimates and the visual side of the program makes it easier to track formulas, R enables to fully leverage its computational power when running a large number of simulations, as will be seen in the following pages.

Finally, for the simulation itself, we define a function that generates a set of random inputs for DCF analysis, using different probability distributions (see previous chapter) to calculate the enterprise value of the firm. We run the MC iteration 10 000 times, which is a good compromise between including too little and too many iterations. The resulting distribution is then stored in the variable output distribution, which is later used to compute other statistics of the distribution, such as mean, standard deviation or deciles.

6.4 Compiling different scenarios probabilities

As already introduced, four different scenarios are being worked on, and each of them has a different probability of occurrence assigned to them. It should be noted that the mean value of the normal distribution differs for each of the four scenarios. Since the scenarios are considered to be normally distributed (and the same standard deviation is used), Monte Carlo simulation can be performed for each of them. The result is a vector of 10,000 values for each scenario, based on the mean value. Then, by considering the assigned probabilities for each scenario and using weighted average, the final sampling distribution is obtained.

The range of potential outcomes for a given revenue growth scenario is shown in the summarizing table below. Specifically, a higher probability of achieving a positive outcome is suggested by the Upside case scenario, with a mean value of €335.9 million and an upper limit of €544.4 million, indicating a potential upside of over €200 million. On the other hand, the most negative outcome is represented by the Worst case scenario, with a mean value of €201.9 million and an upper limit of €322.3 million, implying a potential downside of over €100 million compared to the base case. The Base case scenario represents the most likely outcome, with a mean value of €299.6 million and a balanced range of potential

outcomes from €94.9 million to €484.5 million. Finally, a lower probability of achieving a positive outcome is suggested by the Downside case scenario, with a mean value of €285.1 million and an upper limit of €460.1 million, revealing a potential downside of over €200 million. Now, each of the different scenarios will be shortly introduced with respect to their distributions.

Tab. 6.3 Confidence intervals (95%) of different scenarios (own calculations)

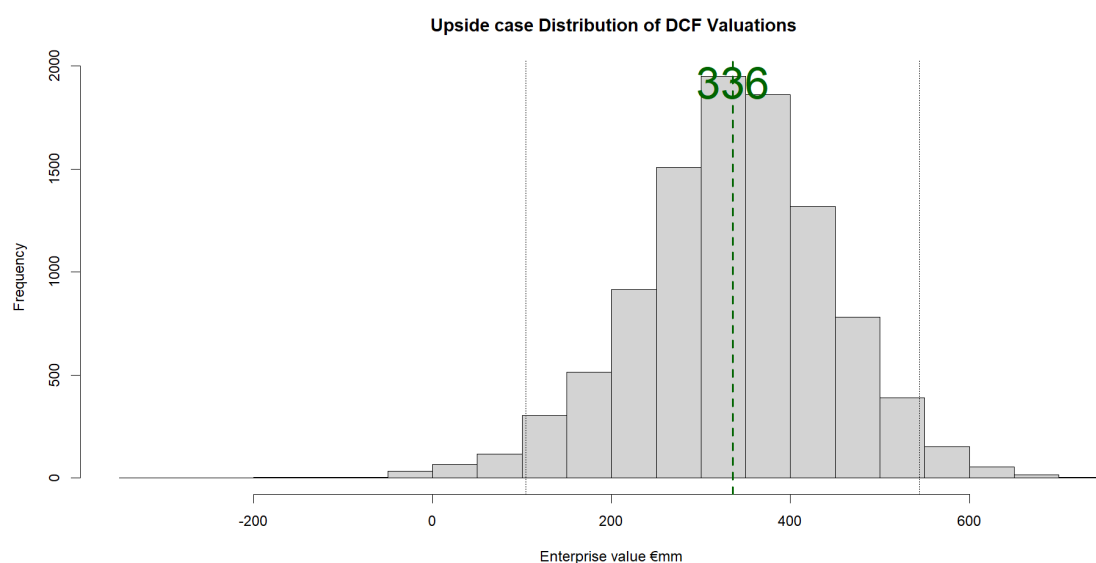
Scenario	Mean value €mm	Lower limit €mm	Upper limit €mm
Upside case	335.9	104.6	544.4
Base case	299.6	94.9	484.5
Downside case	285.1	91.2	460.1
Worst case	201.9	68.6	322.3

6.4.1 Upside case distribution

With a mean value of €335.9 million and an upper limit of €544.4 million, the upside case scenario suggests the potential for significant growth beyond the base case scenario if the company's expansion and growth strategy is successful. However, the probability of this scenario being successful is estimated to be just 10%, which is not very significant.

In the obtained statistics from this particular distribution, some observations can be made. The mean value of 335.9 and the median of 338.9 are quite close, which suggests that the distribution is approximately symmetric. Moreover, the range between the minimum observed value of -321.2 and the maximum value of 743.5 is quite large, indicating that the data is spread out considerably.

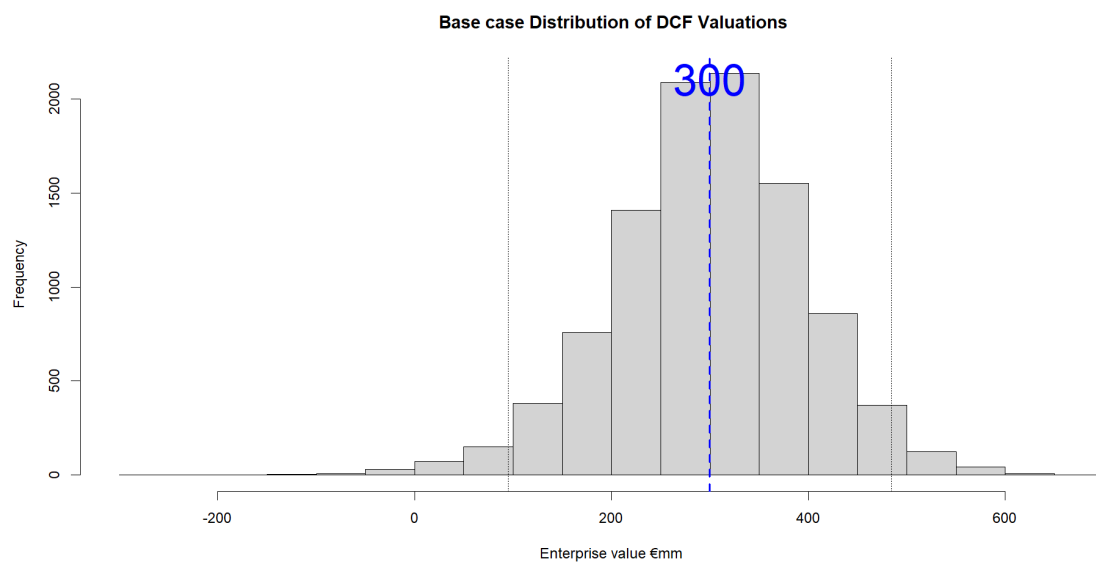
Fig. 6.1 Monte Carlo simulation – Upside case (author's construction)



6.4.2 Base case distribution

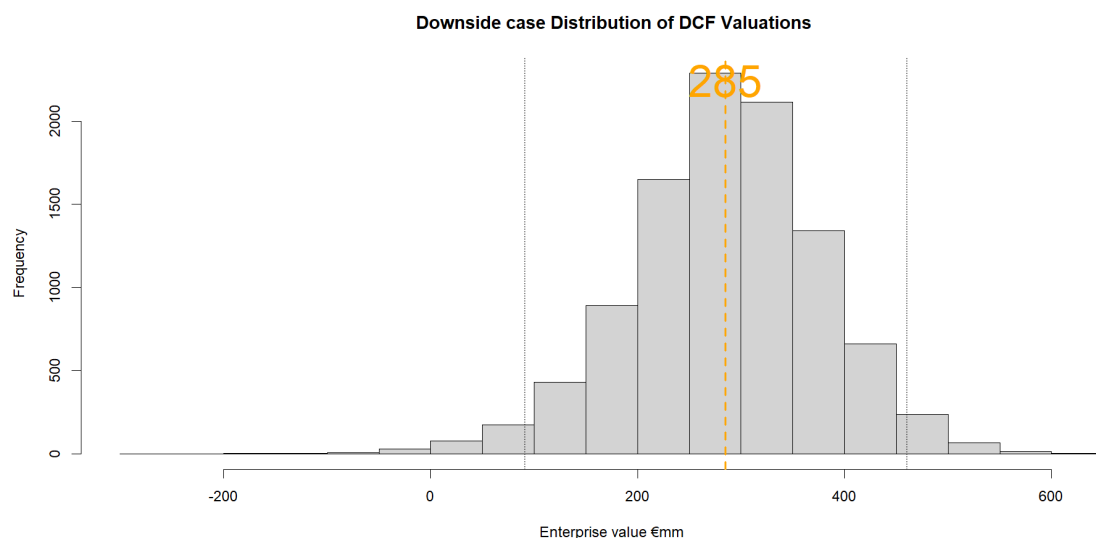
The Base case scenario assumes that the ability of SKODA Auto to slightly outperform the overall market growth rate for passenger cars and light commercial vehicles will be enabled by the company's investment in e-mobility and digitalization, resulting in 1.5% annual growth. While the Base case scenario represents the most likely outcome (based on previous assumptions), there are definitely some risks which can strongly decay revenue growth, more specifically mentioned in downside and worst case scenarios in the chapter 5. Since all scenarios are assumed to have equal distribution with the only difference being the mean, the distributions and their histograms all exhibit very similar characteristics.

Fig. 6.2 Monte Carlo simulation – Base case (author's construction)



6.4.3 Downside case distribution

Fig. 6.3 Monte Carlo simulation – Downside case (author's construction)

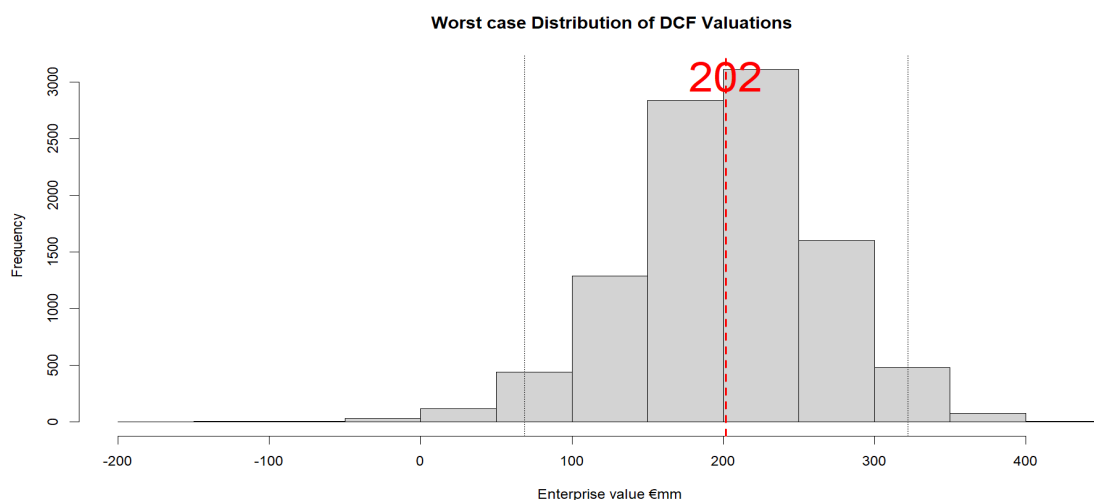


As already mentioned, the simulations resulted in similar distributions with only slight differences in the mean values, as the only varying factor was the revenue growth.

6.4.4 Worst case distribution

While the valuation of SKODA could be affected by an unexpected drop in revenue growth, the Worst case scenario shows it is still far from being a disaster. As was learned in the sensitivity chapter, revenue growth may not be the crucial factor in the overall valuation. Therefore, creating more complex scenarios with varying revenue growth rates and margins can provide a clearer picture of how a company's valuation may be affected by unexpected events, such as black swan events.

Fig. 6.4 Monte Carlo simulation – Worst case (author's construction)



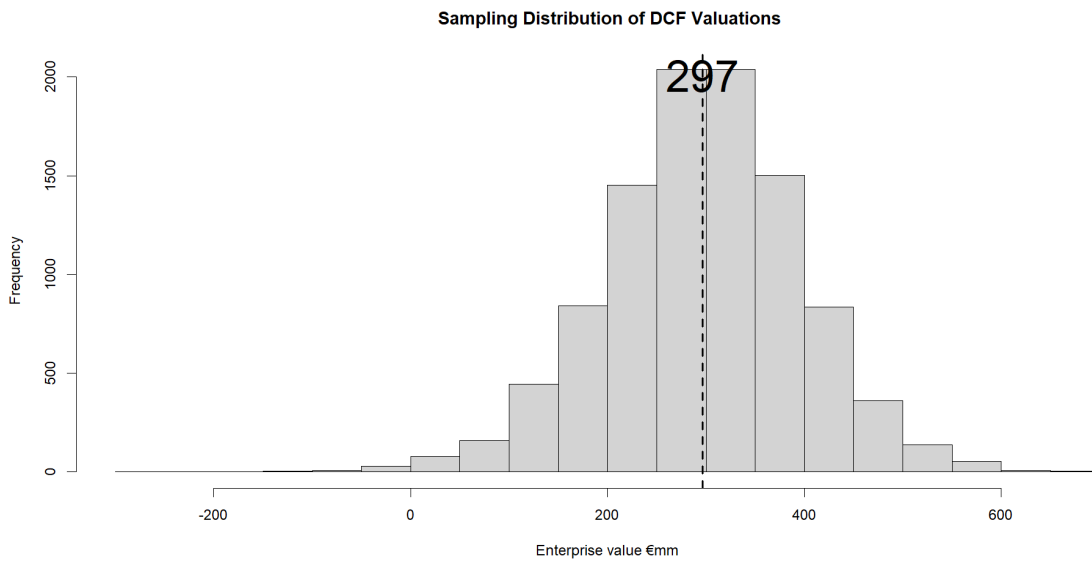
6.5 Monte Carlo sampling distribution

The generated weighted average outputs are presented as frequencies in the histogram below. In the graph, the mean value of 297 €mm can be observed, which is close to the point estimate computed in the beginning (in the deterministic version). The fact that the mean value does not equal the point estimate is not surprising since not all input variables are normally distributed (and therefore are skewed in some direction).

Here we arrive at the key point of the entire analysis. The confidence intervals are likely the most crucial aspect of interest. Firstly, the distribution is estimated based on the sampling distribution, using the Gaussian kernel estimator as discussed in chapter 3.4. The distribution function appears reasonably normal, albeit not completely symmetrical.

In the following lines, the traits of the sampling distribution are calculated and briefly commented on, more specifically measures of central tendency, measures of variability, and measures of shape.

Fig. 6.5 Monte Carlo simulation Sampling distribution (author's construction)



6.5.1 Measures of central tendency

Based on the output distribution from the simulation, here are some thoughts on what the measures of central tendency might imply.

Overall, it appears that the distribution is slightly skewed to the left, as the mean is lower than the median and there is a larger range of values above the median than below the median. This is likely caused by including distributions other than the normal distribution for the model input variables, particularly the log-normal distributions, which are skewed in some direction. Additionally, the distribution has a negative minimum value, although it is not shown in the graph.

Tab. 6.4 Measures of central tendency of the sampling distribution (author's calculation)

Measures of central tendency	Value (€mm)
Min	(280.9)
1 st Quartile	235.1
Mean	296.6
Median	298.6
3 rd Quartile	361.4
Max	681.4

6.5.2 Measures of the variability

The presence of a large range in the distribution suggests that extreme values were possibly obtained from the simulation. It is already known that the output distribution includes negative values, which is not surprising. This is caused by the specifications of the input variables that allowed for negative values in some cases, such as NWC. Even though the probability of obtaining negative EV is extremely small, some negative values were still obtained due to the large number of iterations.

The spread of the values around the mean value can be indicated by the variance and standard deviation. The relatively large standard deviation in our distribution only confirms the large range of values mentioned in the previous paragraph.

On the other hand, information about how the values are dispersed between the 3rd and 1st quartile (i.e. the middle 50% of the values) is provided by the IQR (Interquartile range). In comparison to the large variance and range, the relatively small IQR suggests that many of the values are clustered around the median. This indicates that the distribution may have relatively high kurtosis, which will be explored further in the upcoming chapter.

Tab. 6.5 Measures of variability of the sampling distribution (author's calculation)

Measures of variability	
Range	962.3
Variance	10 038.4
Standard deviation	100.2
IQR (Interquartile Range)	126.3

6.5.3 Measures of the shape

Measures of the shape help us to further understand the overall shape of the distribution. Starting with kurtosis, the value of 3.7 confirms that the distribution is definitely more peaked than the normal distribution, meaning it is leptokurtic. We can further assume that the tails of the distribution are quite thick, while the peak is quite narrow.

The degree of skewness being small and not too far from zero suggests that the distribution is approximately symmetrical, but not perfectly symmetrical. It tends to have more values clustered to the right side, indicating a slightly negative skew. However, we should also consider that skewness is relatively sensitive to outliers, which could have an impact on the value.

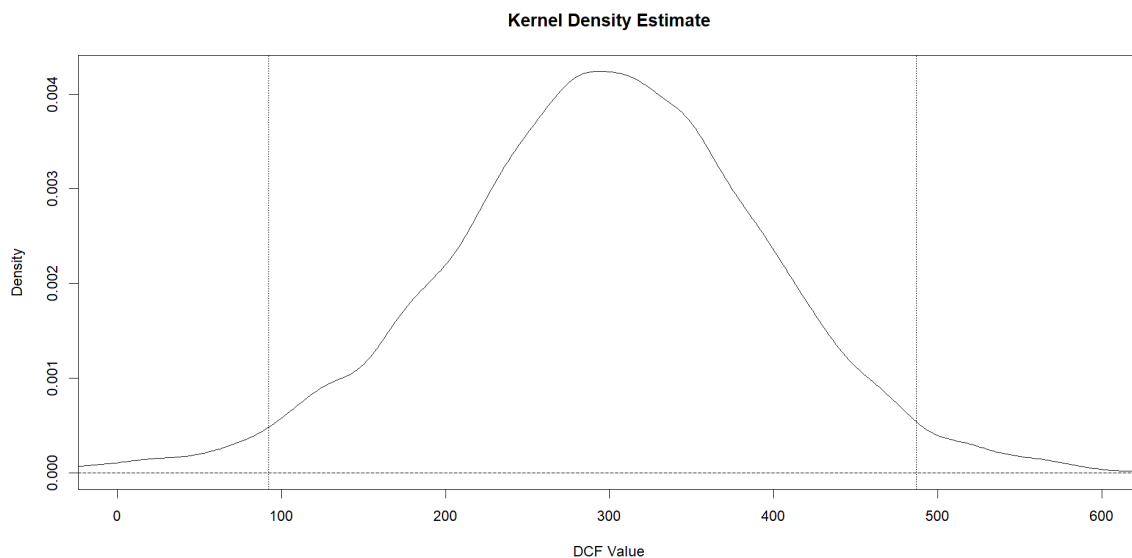
Tab. 6.6 Measures of the shape of the sampling distribution (author's calculation)

Measures of the shape	
Skewness	(0.21)
Kurtosis	3.71

6.5.4 Enterprise value confidence intervals

After a Monte Carlo extension of the DCF model was conducted to account for input variability, a 95% confidence interval was obtained with a range from 92 €mm to 487 €mm. This wide range highlights the significant amount of variability present in the model inputs and demonstrates the challenge of obtaining a single point valuation estimate.

Fig. 6.6 Using Kernel estimator to demonstrate the confidence intervals (author's construction)



Incorporating Monte Carlo simulation to factor in the input variability seems to be a good approach as it resulted in a more practical model that better captures the inherent uncertainty in the company valuation. As a result, we gained valuable insights into potential valuation outcomes under different scenarios and more importantly, under different probabilities.

Overall, the incorporation of Monte Carlo simulation into DCF valuation models can help to develop a more comprehensive view of company valuations that considers the inherent variability in the investment landscape. This can ultimately help us make more informed and data-driven investment decisions, which are grounded in a deeper understanding of the potential risks and uncertainties associated with different investment opportunities.

7 Sensitivity analysis

Sensitivity analysis is a crucial part of every financial model as it helps us evaluate how changes in key assumptions and variables impact the model's outcomes. By systematically varying inputs and analysing the resulting outputs, sensitivity analysis can help identify the most important drivers of value and highlight areas of uncertainty and risk. For my model, I chose to test four key variables that I think have the most impact on the final Enterprise value:

- Revenue growth
- EBITDA margin
- Cost of Capital (WACC)
- Terminal growth rate

Each of the variables is then tested for sensitivity while other variables are held constant. By doing so, we can ensure that no other effects are projected in the final value. In the table below, we can see the ranges in which we are going to test the key drivers. The blue-coded percentages represent the original input assumptions, while the black values represent the options to be tested. Each variable has a slightly different range of tested values because of the different nature of sensitivity. For instance, a one percentage point change in Terminal growth assumptions has a completely different effect on the Enterprise value than a percentage point change in revenue growth. Therefore, each range has been selected based on this preliminary idea.

The figures were exported to Excel because it was deemed better to work with data that could be easily viewed for this type of analysis. Additionally, creating a complex sensitivity analysis is more straightforward in Excel, which led to the decision to export the figures.

Tab. 7.1 Sensitivity table ranges of key variables (author's calculation)

Drivers	Sensitivity range				
Revenue growth	0.50%	1.00%	1.50%	2.00%	2.50%
EBITDA margin	9.30%	10.80%	12.30%	13.80%	15.30%
WACC	7.80%	8.80%	9.80%	10.80%	11.80%
Terminal growth	0.00%	0.25%	0.50%	0.75%	1.00%

With the sensitivity ranges used, the complexity of the model was tested, resulting in the outcomes presented below. It can be observed that the average value always falls in the middle of the sensitivity table for each parameter.

Also, a max/min chart was created by taking the extremes from each of the sensitivity tables to obtain a better overview of the potential upswings and downswings. Additionally, these charts are useful for preparing the data for the graphical exercise.

Tab. 7.2 Sensitivity analysis results (author's calculation)

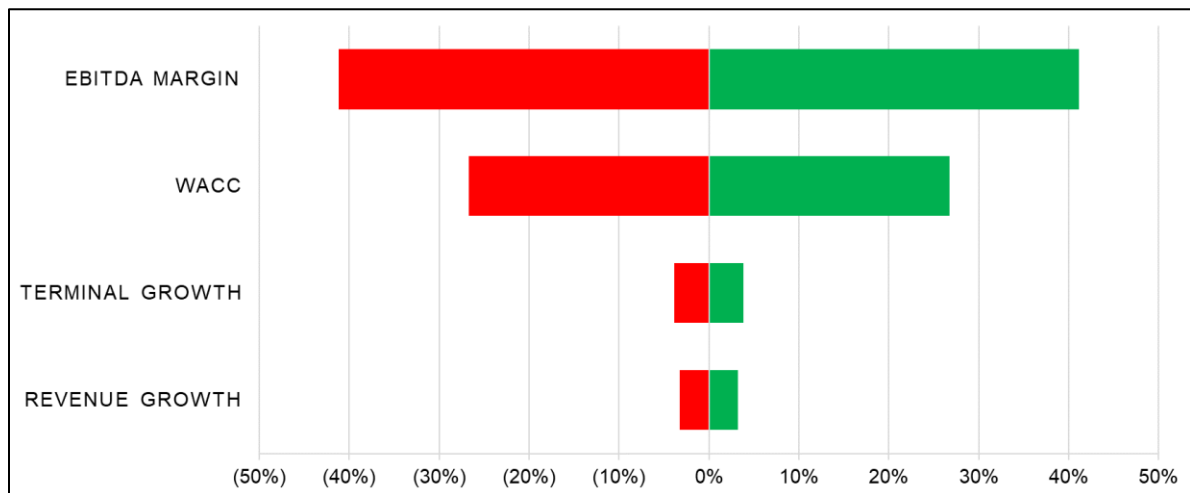
Revenue growth	EV €mm		Revenue growth +/- 1%	
	0.50%	291.4	Change	EV €mm
	1.00%	296.3	0.00%	301.2
	1.50%	301.2	(1.00%)	291.4
	2.00%	306.2	1.00%	311.3
	2.50%	311.3		
EBITDA margin	EV €mm		EBITDA margin +/- 3%	
	9.30%	177.3	Change	EV €mm
	10.80%	239.3	0.00%	301.2
	12.30%	301.2	(3.00%)	177.3
	13.80%	363.1	3.00%	425.0
	15.30%	425.0		
WACC	EV €mm		WACC +/- 2%	
	7.75%	381.7	Change	EV €mm
	8.75%	335.5	0.00%	301.2
	9.75%	301.2	(2.00%)	381.7
	10.75%	270.7	2.00%	247.0
	11.75%	247.0		
Terminal growth	EV €mm		Terminal growth +/- .5%	
	0.00%	289.6	Change	EV €mm
	0.25%	295.3	0.00%	301.2
	0.50%	301.2	(0.50%)	289.6
	0.75%	307.5	0.50%	314.1
	1.00%	314.1		

The ultimate output of this sensitivity analysis comes in the form of graph, more specifically so-called tornado chart. Tornado charts are a type of bar chart that reflect how much impact varying an input has on a particular output, providing both a ranking and a measure of magnitude of the impact, sometimes given in absolute terms and sometimes in percentage terms. (Bastick, 2023) The percentage terms are used for analysis in this study, although the change is calculated in both absolute and percentage terms, as shown in the sensitivity tables above.

These charts show the results of single factor sensitivity analysis, i.e. outcomes displayed by changing one variable at a time. The variables are positioned on the chart vertically, rather than horizontally, in such a way that the largest swing appears at the top of the chart, the second largest appears second from the top and so on, with the lowest swing appearing at the bottom of the chart. Hence, the final chart visually resembles either one half of, or a complete, tornado. (Anon., 2019)

Furthermore, it should be noted that the variables are tested symmetrically, meaning that the range of tested values is equally balanced around the actual mean value of the tested variables. If we had more detailed data and insights directly from the company, we would be able to better assess the possible ranges of the outcomes. Therefore, the resulting tornado chart might be less symmetrical due to multiple factors such as historical patterns, insider information, etc.

Fig. 7.1 Tornado sensitivity chart of key variables (author's construction)



The most sensitive variable in our model is the EBITDA margin, which has a significant impact on the total Enterprise value, with a potential change (either positive or negative) of more than 40% by a mere 3 percentage points. Similarly, the Cost of Capital, WACC, has a notable effect on the Terminal value, with a potential variability of almost 27% if altered by 2 percentage points.

Interestingly, the Revenue growth and Terminal growth did not demonstrate the same level of impact as initially anticipated. For instance, a change of 1 percentage point in revenue growth resulted in only a 3.3% variability in the outcome. Similarly, a half-percentage point alteration in the Terminal growth produced a 3.8% variability on both ends of the enterprise value.

Conclusion

The main objective of this thesis was to value SKODA using the DCF method and subsequently obtain the sampling distribution of enterprise value probabilities using Monte Carlo simulation, along with an examination of different scenarios and their potential outcomes.

Overall, the results of this thesis demonstrate that incorporating the Monte Carlo simulation into the DCF valuation proved that it could increase the reliability of the results as evidenced by the sampling distribution and confidence intervals derived from the simulation.

Although Monte Carlo might be a great extension of DCF model, it also comes with certain limitations. The input variables require assigning probability distributions which can be quite challenging and not so straightforward. Some might argue that using the past financials gives us a clear idea of how distributed the variables should be. However, this approach would require extremely long time-series of financials and still one cannot be sure how the future will look like for the firm. This fact leaves a room for potential bias towards some assumptions when it comes to determining the right probability distribution of variables. Therefore, it is of great significance to think of all the important factors which might affect the distribution and to choose the ones which have the most significant impact.

The impact of using a log-normal distribution for some variables on the overall sampling distribution was observed in this work. If a normal distribution was used exclusively, the central limit theorem would have ensured that the distribution would become almost normal after running the simulation 10,000 times. However, this was not observed, and a slightly right-skewed distribution resulted.

The deterministic DCF model gives us Enterprise value of 311 €mm as of end of September 2022. Since we do not know what the market value is because of SKODA not being public company, we unfortunately cannot check with any consensus. However, the EV/EBITDA multiple of 5.6x (slight decrease from 5.9x in 2021) serves as a good sanity check that the derived value is not completely off the scale. Moreover, after conducting the Monte Carlo simulation and determining the 95% enterprise value confidence interval to lie between 92 €mm (1.7x EV/EBITDA) and 487 €mm (8.8x EV/EBITDA), we can be almost sure that the Enterprise value would actually fall within this range.

In the end we conducted a sensitivity analysis of four key variables of the model which helped us to identify EBITDA margin and WACC as the key drivers of the Enterprise value. This sensitivity analysis came quite surprising as the initial hypothesis was that the Revenue growth would be among the key drivers. Therefore, the initial hypothesis proved to be wrong.

Overall, extending the DCF model with Monte Carlo simulation is a potent technique that can provide a more accurate and practical understanding of valuation risks and

opportunities. I believe that this approach could be a useful tool for anyone seeking to develop more thorough and precise financial models.

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